

Universality of subshifts of finite type for groups

Sebastián Barbieri

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Abstract

These are notes meant to complement a three-lecture minicourse exploring the phenomenon of self-simulation on groups. They are not meant to be a full description of the topic but only to serve as a reference to the interested reader. These notes were drafted using Claude based on the existing literature and were then edited extensively by the author.

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1 Lecture I: shift spaces and computability

Throughout these notes G denotes a countable group, written multiplicatively, with identity element 1_G , and Σ denotes a finite set with at least two elements. We call Σ the *alphabet* and its elements *symbols*. From [Section 1.6](#) onwards we shall further assume that G is finitely generated, but for now we may proceed in general.

In topological dynamics, we are interested in studying group actions $G \curvearrowright^\alpha X$, where X is a compact metrizable space and $\alpha: G \times X \rightarrow X$ is an action by homeomorphisms of X . In these notes we will focus on the special case where X has zero topological dimension, and thus we can identify it with a closed subset of the Cantor space $\{0, 1\}^{\mathbb{N}}$.

The material of this section is classical. For the case $G = \mathbb{Z}$ we refer to the book of Lind and Marcus [LM21]. For some parts of the general group setting we refer to Ceccherini-Silberstein and Coornaert [CSC10], and for some computability aspects to the survey [ABJ18].

1.1 The full shift

Definition 1.1 (Configurations and the full shift). A *configuration* is a map $x: G \rightarrow \Sigma$. We write x_g for the value $x(g)$ and think of x as a coloring of the group by elements of Σ . The set of all configurations

$$\Sigma^G = \{x: G \rightarrow \Sigma\}$$

is called the *full shift*. It carries the *prodiscrete topology*, that is, the product topology obtained by equipping each copy of Σ with the discrete topology.

We write $F \Subset G$ to mean that F is a finite subset of G . Given $F \Subset G$, a map $p \in \Sigma^F$ is called a *pattern* with support F . Given $x \in \Sigma^G$ and $F \subseteq G$, we denote the pattern induced by its restriction to F by $x|_F \in \Sigma^F$.

A basis of the prodiscrete topology is given by the sets obtained by prescribing finitely many coordinates. In other words, the cylinder sets of configurations that are obtained by enforcing that a specific pattern occurs.

Definition 1.2 (Cylinders). Let $F \Subset G$ and $p \in \Sigma^F$. The associated *cylinder* is

$$[p] = \{x \in \Sigma^G : x|_F = p\}.$$

More generally, for $g \in G$ we write $[p]_g = \{x : x_{gf} = p_f \text{ for every } f \in F\}$, so that $[p] = [p]_{1_G}$.

Cylinders are both open and closed: $[p]$ is open by definition of the product topology, and its complement is the finite union of the cylinders $[p']$ with $p' \in \Sigma^F \setminus \{p\}$, which is open as well. Consequently Σ^G is totally disconnected.

Proposition 1.3. *The space Σ^G is compact, totally disconnected and metrizable. If moreover G is infinite and $|\Sigma| \geq 2$, then Σ^G is homeomorphic to the Cantor set.*

Proof sketch. Compactness follows immediately by Tychonoff's theorem, and total disconnectedness was observed above. For metrizability, fix an enumeration $(g_i)_{i \geq 1}$ of the group elements and let

$$d(x, y) = \begin{cases} 2^{-\min\{n \geq 1 : x(g_n) \neq y(g_n)\}} & \text{if } x \neq y, \\ 0 & \text{if } x = y. \end{cases}$$

It can easily be checked that d is an ultrametric which induces the prodiscrete topology. Finally, a nonempty compact metrizable totally disconnected space without isolated points is homeomorphic to the Cantor set, by Brouwer's characterization. Furthermore, no point of Σ^G is isolated when G is infinite and $|\Sigma| \geq 2$. $\square$

Next we define the shift action.

Definition 1.4 (shift action). The left *shift action* of $G \curvearrowright^\sigma \Sigma^G$ is the map $\sigma: G \times \Sigma^G \rightarrow \Sigma^G$ given by

$$\sigma^g(x)_h = x_{g^{-1}h}, \quad g, h \in G, \quad x \in \Sigma^G.$$

In most cases where there is no possible confusion, we shall write $g \cdot x$ instead of $\sigma^g(x)$ to simplify the notation.

Lemma 1.5. $G \curvearrowright \Sigma^G$ is a left action of G on Σ^G by homeomorphisms.

Proof. For $g_1, g_2, h \in G$ we compute

$$\sigma^{g_1}(\sigma^{g_2}(x))_h = \sigma^{g_2}(x)_{g_1^{-1}h} = x_{g_2^{-1}g_1^{-1}h} = x_{(g_1g_2)^{-1}h} = \sigma^{g_1g_2}(x)_h,$$

and $\sigma^{1_G} = \text{id}$. Each σ^g is thus a bijection with inverse $\sigma^{g^{-1}}$. Continuity follows from the identity $(\sigma^g)^{-1}([q]) = [q]_{g^{-1}}$ for every cylinder $[q]$: preimages of basic open sets are cylinders, hence open. $\square$

Remark 1.6. The inverse in Definition 1.4 is what makes σ a left action. Some authors prefer the left action by right multiplication $(g \cdot x)_h = x_{hg}$, or one of the two possible right actions $(x \cdot g)_h = x_{gh}$ or $(x \cdot g)_h = x_{hg^{-1}}$. Everything below can be translated between distinct conventions, but the translation must be done consistently; mixing them is a frequent source of errors when G is not abelian (trust me, I once wrote a 40 page paper that had to be thrown into the trash due to an error like that!).

Example 1.7 ($G = \mathbb{Z}$). Here a configuration is a bi-infinite sequence $x = (x_n)_{n \in \mathbb{Z}}$ of symbols, and $\sigma^1(x)_n = x_{n-1}$, so that σ^1 shifts the sequence one step to the right. With $\Sigma = \{0, 1\}$ the full shift $\{0, 1\}^{\mathbb{Z}}$ is the classical binary full shift.

Example 1.8 ($G = \mathbb{Z}^2$). A configuration is a coloring of the integer grid, and $\sigma^{(m,n)}$ translates the coloring by the vector (m, n) . Pictures of elements of $\Sigma^{\mathbb{Z}^2}$ are the usual way of representing these objects.

Example 1.9 ($G = \mathbb{F}_2$). Let $\mathbb{F}_2 = \langle a, b \rangle$ be the free group of rank 2. Its Cayley graph with respect to $\{a^{\pm 1}, b^{\pm 1}\}$ is the 4-regular tree, and a configuration is a coloring of the vertices of that tree. Already here the geometry of G differs drastically from the $\mathbb{Z}^d$ case: balls grow exponentially.

Example 1.10 (G finite). If G is finite then Σ^G is a finite discrete space. The theory below is not interesting in this case, and the reader may safely assume G infinite throughout.

1.2 Shift spaces

Let $p \in \Sigma^F$ be a pattern and $x \in \Sigma^G$. We say that p occurs in x at position $g \in G$ if

$$x_{gf} = p_f \quad \text{for every } f \in F,$$

equivalently if $g^{-1} \cdot x \in [p]$. We say that p occurs in x if it occurs at some position, and write $p \sqsubseteq x$.

Definition 1.11 (Subshifts). Let $\mathcal{F}$ be a set of patterns, which we call *forbidden*. Define

$$X_{\mathcal{F}} = \{x \in \Sigma^G : p \not\sqsubseteq x \text{ for every } p \in \mathcal{F}\} = \Sigma^G \setminus \bigcup_{p \in \mathcal{F}} \bigcup_{g \in G} [p]_g.$$

A subset $X \subseteq \Sigma^G$ is a *subshift* (or a *shift space*) if $X = X_{\mathcal{F}}$ for some $\mathcal{F}$.

Proposition 1.12. A subset $X \subseteq \Sigma^G$ is a subshift if and only if it is closed and invariant under the shift action, that is, $\sigma^g(X) = X$ for every $g \in G$.

Proof. Suppose $X = X_{\mathcal{F}}$. We have that X is the complement of a union of cylinders, hence closed. If $x \in X$ and $h \in G$, then a pattern p occurs in $h \cdot x$ at position g if and only if it occurs in x at position $h^{-1}g$; since $g \mapsto h^{-1}g$ is a bijection of G , the configurations x and $h \cdot x$ contain exactly the same patterns, so $h \cdot x \in X$.

Conversely, suppose X is closed and invariant, and let

$$\mathcal{F} = \{p \in \bigcup_{F \in G} \Sigma^F : p \not\subseteq x \text{ for every } x \in X\}$$

be the set of patterns occurring in no element of X . Clearly $X \subseteq X_{\mathcal{F}}$. For the reverse inclusion let $x \notin X$. As X is closed, its complement is open, so there are $F \in G$ and a cylinder with $x \in [x|_F] \subseteq \Sigma^G \setminus X$. We claim that the pattern $q = x|_F$ occurs in no element of X . Indeed, if q occurred in some $y \in X$ at a position $g \in G$, then $g^{-1} \cdot y \in [q]$, while $g^{-1} \cdot y \in X$ by invariance, contradicting that $[q]$ is disjoint from X . Hence $q \in \mathcal{F}$ and, as q occurs in x , we get $x \notin X_{\mathcal{F}}$. $\square$

Remark 1.13. The set $\mathcal{F}$ is by no means unique: the same subshift may be described by many different sets of forbidden patterns.

Example 1.14 (Extreme cases). Taking $\mathcal{F} = \emptyset$ gives $X_{\mathcal{F}} = \Sigma^G$, and taking $\mathcal{F} = \Sigma^{\{1_G\}}$, that is, forbidding every single letter, gives $X_{\mathcal{F}} = \emptyset$. Both the full shift and the empty set are subshifts.

Definition 1.15 (Language). Let $X \subseteq \Sigma^G$ be a subshift. For $F \in G$ put

$$\mathcal{L}_F(X) = \{x|_F : x \in X\} \subseteq \Sigma^F, \quad \mathcal{L}(X) = \bigcup_{F \in G} \mathcal{L}_F(X).$$

The set $\mathcal{L}(X)$ is the *language* of X , and its elements are the *globally admissible* patterns of X . Notice that every subshift can be defined by taking $\mathcal{F}$ as the complement of $\mathcal{L}(X)$.

Example 1.16 (Golden mean shift). Let $G = \mathbb{Z}$, $\Sigma = \{0, 1\}$ and $\mathcal{F} = \{11\}$, where 11 denotes the pattern with support $\{0, 1\}$ and constant value 1. The SFT $X_{\mathcal{F}}$ consists of all binary sequences with no two consecutive 1's. Its language is the set of finite binary words avoiding 11, whose cardinality grows like the Fibonacci numbers, hence the name.

Example 1.17 (Hard square shift). Let $G = \mathbb{Z}^2$, $\Sigma = \{0, 1\}$ and forbid the two patterns of constant value 1 with supports $\{(0, 0), (1, 0)\}$ and $\{(0, 0), (0, 1)\}$. The resulting SFT consists of the colorings of the grid in which the set of 1's is an independent set of the grid graph. This is the two-dimensional analogue of the previous example, and almost nothing about it is as simple as in dimension one.

Example 1.18 (Wang tiles). Wang tiles were introduced in [Wan61]. A *Wang tile* is a unit square with a color on each side, formally an element $\tau = (\tau_N, \tau_E, \tau_S, \tau_W) \in C^4$ for a finite set of colors C . Given a finite set $T \subseteq C^4$ of Wang tiles, the associated *Wang shift* is

$$X_T = \{x \in T^{\mathbb{Z}^2} : (x_v)_E = (x_{v+(1,0)})_W \text{ and } (x_v)_N = (x_{v+(0,1)})_S \text{ for all } v \in \mathbb{Z}^2\},$$

that is, the set of tilings of the plane by translated copies of the tiles of T in which adjacent tiles share the color of their common edge, see Figure 1.

Example 1.19 (Sunny-side up). Consider the alphabet $\Sigma = \{0, 1\}$ and the *sunny-side up* subshift

$$X_{\leq 1} = \{x \in \{0, 1\}^G : \#\{g \in G : x_g = 1\} \leq 1\}.$$

This subshift can be thought of as an encoding of the natural action of G on its one-point compactification $G \cup \{\infty\}$.

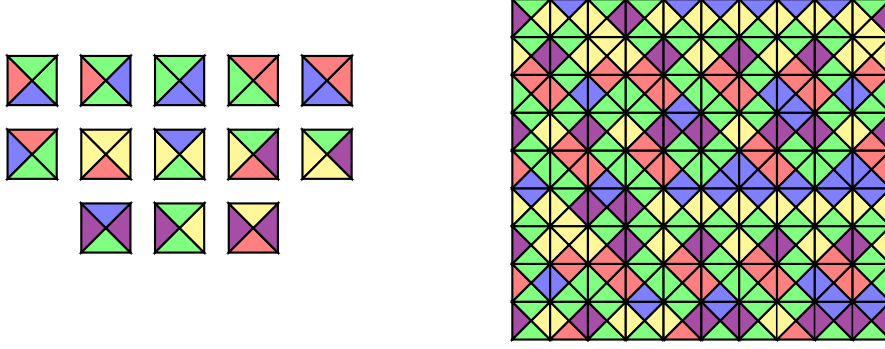


Figure 1: On the left a finite alphabet of Wang tiles, on the right a finite portion of an element of the associated shift space, which is defined as the subset of all maps from $\mathbb{Z}^2$ to the alphabet where adjacent squares share the same color. This particular shift space is the well-known Kari–Culik tiling [Kar96, Cul96], which realizes a free $\mathbb{Z}^2$ -action using the 13 tiles of [Cul96].

1.3 Morphisms between dynamical systems

Consider two actions by homeomorphisms $G \curvearrowright^\alpha X$ and $G \curvearrowright^\beta Y$, where X and Y are compact metrizable spaces.

Definition 1.20. A *topological morphism* between $G \curvearrowright^\alpha X$ and $G \curvearrowright^\beta Y$ is a continuous map $\phi: X \rightarrow Y$ which is G -equivariant, that is, such that for every $x \in X$ it satisfies $\phi(g \cdot x) = g \cdot \phi(x)$. Furthermore, it is called

1. a *factor map* if ϕ is surjective. In this case we say $G \curvearrowright^\beta Y$ is a *factor* of $G \curvearrowright^\alpha X$, and that $G \curvearrowright^\alpha X$ is an *extension* of $G \curvearrowright^\beta Y$.
2. a *topological conjugacy* if ϕ is a bijection. In this case we say $G \curvearrowright^\alpha X$ and $G \curvearrowright^\beta Y$ are *conjugate*.

Notice that as X is compact and Y is Hausdorff, the inverse of a bijective morphism is automatically continuous, so every conjugacy is automatically a homeomorphism. The notion of topological conjugacy is the right notion of isomorphism for group actions by homeomorphisms. Next we characterize shift spaces through the notion of expansivity.

Definition 1.21. Let d be a compatible metric on X . We say an action by homeomorphisms $G \curvearrowright X$ is *expansive* if there exists $C > 0$ such that whenever $d(g \cdot x, g \cdot y) < C$ for all $g \in G$, then $x = y$.

In simpler terms, an action is expansive if every pair of distinct points is eventually “far” apart along their orbits.

Theorem 1.22. An action $G \curvearrowright X$ is topologically conjugate to a subshift if and only if X is zero-dimensional and $G \curvearrowright X$ is expansive.

Proof. Suppose first that $G \curvearrowright X$ is conjugate to a subshift; as both conditions are preserved by conjugacy we may assume that $X \subseteq \Sigma^G$ is itself a subshift. Being a subspace of Σ^G , it is zero-dimensional by Proposition 1.3. For expansivity, equip X with the metric d from the proof of Proposition 1.3, choosing the enumeration of G so that $g_1 = 1_G$, and let $C = 1$. By the definition of d , two configurations satisfy $d(u, v) < 1$ precisely when $u_{1_G} = v_{1_G}$. Hence if $d(g \cdot x, g \cdot y) < C$ for every $g \in G$, then

$$x_{g^{-1}} = (g \cdot x)_{1_G} = (g \cdot y)_{1_G} = y_{g^{-1}} \quad \text{for every } g \in G,$$

and as $g \mapsto g^{-1}$ is a bijection of G we conclude $x = y$.

Conversely, let X be zero-dimensional and let $C > 0$ be an expansivity constant for a compatible metric d . As X is compact, metrizable and zero-dimensional, it admits a basis of clopen sets, so every point lies in a clopen set of diameter less than C . By compactness finitely many of them, say $U_1, \dots, U_k$, cover X . Setting

$$P_1 = U_1, \quad P_i = U_i \setminus (U_1 \cup \dots \cup U_{i-1}) \quad (2 \leq i \leq k),$$

we obtain a finite partition $\{P_1, \dots, P_k\}$ of X into clopen sets, each of diameter less than C (empty atoms may be discarded).

Let $\Sigma = \{1, \dots, k\}$ and define $\varphi: X \rightarrow \Sigma^G$ by

$$\varphi(x)_g = i \quad \text{if and only if} \quad g^{-1} \cdot x \in P_i,$$

which makes sense because the atoms partition X . We claim that φ is a conjugacy onto its image.

Equivariance. For $h, g \in G$ we have $\varphi(h \cdot x)_g = i$ if and only if $g^{-1}h \cdot x \in P_i$, while $(h \cdot \varphi(x))_g = \varphi(x)_{h^{-1}g} = i$ if and only if $(h^{-1}g)^{-1} \cdot x = g^{-1}h \cdot x \in P_i$. Hence $\varphi(h \cdot x) = h \cdot \varphi(x)$.

Continuity. Let $F \subseteq G$ and $a \in \Sigma^F$. Then $\varphi(x)|_F = a$ if and only if $g^{-1} \cdot x \in P_{a_g}$, that is $x \in g \cdot P_{a_g}$, for every $g \in F$. Therefore

$$\varphi^{-1}([a]) = \bigcap_{g \in F} g \cdot P_{a_g},$$

a finite intersection of clopen sets, and in particular an open set. As the cylinders form a basis, φ is continuous.

Injectivity. Suppose $\varphi(x) = \varphi(y)$. Then for every $g \in G$ the points $g^{-1} \cdot x$ and $g^{-1} \cdot y$ lie in the same atom, whence $d(g^{-1} \cdot x, g^{-1} \cdot y) < C$. As g runs over G so does g^{-1} , and expansivity yields $x = y$.

Thus φ is a continuous equivariant injection from a compact space into a Hausdorff space, hence a homeomorphism onto $Y = \varphi(X)$, and this homeomorphism is equivariant. Finally Y is compact, hence closed, and it is shift invariant by equivariance, so Y is a subshift by [Proposition 1.12](#). Therefore $G \curvearrowright X$ is topologically conjugate to the subshift Y . $\square$

Remark 1.23. Expansivity is stated in terms of a compatible metric, but for compact X the property does not depend on which one is chosen: only the constant C does. The proof above also shows where the finiteness of the alphabet comes from, namely from the compactness of X together with the existence of arbitrarily fine clopen partitions.

In the case of subshifts, topological morphisms can be described in combinatorial terms through cellular automata. In what follows Σ and Σ' denote two finite alphabets, and we consider maps between subshifts $X \subseteq \Sigma^G$ and $Y \subseteq \Sigma'^G$.

Definition 1.24 (Sliding block code). Let $F \subseteq G$ and let $\Phi: \Sigma^F \rightarrow \Sigma'$ be any map, called a *local rule* with *window* F . The associated *sliding block code* is

$$\varphi_\Phi: \Sigma^G \rightarrow \Sigma'^G, \quad \varphi_\Phi(x)_g = \Phi((g^{-1} \cdot x)|_F),$$

that is, the letter written at g is computed by the single rule Φ from the letters $(x_{gf})_{f \in F}$ read in the window F translated to g .

A 1-block code is the special case $F = \{1_G\}$. The point of the next theorem is that, up to enlarging the window, there are no other morphisms.

Theorem 1.25 (Curtis–Hedlund–Lyndon). *Let $X \subseteq \Sigma^G$ and $Y \subseteq \Sigma'^G$ be subshifts and let $\varphi: X \rightarrow Y$ be a map. Then φ is a topological morphism if and only if there are $F \in G$ and a local rule $\Phi: \Sigma^F \rightarrow \Sigma'$ such that $\varphi = \varphi_\Phi|_X$.*

Proof. Suppose first that $\varphi = \varphi_\Phi|_X$ for some local rule Φ with window F . Equivariance is a computation: for $h, g \in G$,

$$\varphi_\Phi(h \cdot x)_g = \Phi((g^{-1}h \cdot x)|_F) = \varphi_\Phi(x)_{h^{-1}g} = (h \cdot \varphi_\Phi(x))_g.$$

Continuity holds because the coordinate $\varphi_\Phi(x)_g$ depends only on the finitely many coordinates $(x_{gf})_{f \in F}$: the preimage of the cylinder $[b]$, $b \in \Sigma'$, is the finite union of the cylinders $[q]$ with $q \in \Sigma^F$ and $\Phi(q) = b$, and cylinders of this form generate the topology of Σ'^G under translation.

Conversely, let φ be a morphism. For each letter $b \in \Sigma'$ put

$$U_b = \varphi^{-1}([b] \cap Y) = \{x \in X : \varphi(x)_{1_G} = b\}.$$

The sets U_b are open, since φ is continuous and $[b]$ is open, and they partition X ; being complements of finite unions of open sets, they are also closed. Every open subset of X is a union of cylinders of X , and U_b is closed in the compact space X , hence compact, so finitely many cylinders suffice: there is a finite set of patterns Q_b with $U_b = \bigcup_{q \in Q_b} ([q] \cap X)$. Let

$$F = \bigcup_{b \in \Sigma'} \bigcup_{q \in Q_b} \text{supp}(q) \in G,$$

which is finite because Σ' and each Q_b are. Enlarging every pattern of every Q_b to the support F , that is, replacing it by the finitely many patterns of Σ^F which extend it, we may assume that all of them have support exactly F . Since the U_b partition X , for every $q \in \mathcal{L}_F(X)$ there is exactly one $b \in \Sigma'$ with $[q] \cap X \subseteq U_b$; define $\Phi(q) = b$, and define Φ arbitrarily on $\Sigma^F \setminus \mathcal{L}_F(X)$. By construction

$$\varphi(x)_{1_G} = \Phi(x|_F) \quad \text{for every } x \in X.$$

Finally, equivariance transports this identity to every coordinate: for $x \in X$ and $g \in G$,

$$\varphi(x)_g = \sigma^{g^{-1}}(\varphi(x))_{1_G} = \varphi(\sigma^{g^{-1}}(x))_{1_G} = \Phi(\sigma^{g^{-1}}(x)|_F) = \varphi_\Phi(x)_g,$$

where the second equality uses that φ is a morphism. Hence $\varphi = \varphi_\Phi|_X$. $\square$

Remark 1.26. The theorem is due to Hedlund [Hed69] for $G = \mathbb{Z}$; the proof above is the standard one and works verbatim for an arbitrary group, see [CSC10, Section 1.8]. Compactness of X is essential: it is what turns the a priori infinite amount of information determining $\varphi(x)_{1_G}$ into a rule reading a fixed finite window. The statement fails for infinite alphabets.

1.4 Subshifts of finite type and sofic subshifts

Definition 1.27 (SFT). A subshift $X \subseteq \Sigma^G$ is a *subshift of finite type* (SFT) if $X = X_{\mathcal{F}}$ for some *finite* set $\mathcal{F}$ of forbidden patterns.

Remark 1.28. A subshift $X \subseteq \Sigma^G$ is an SFT if and only if there exist $F \in G$ and $L \subseteq \Sigma^F$ such that

$$x \in X \text{ if and only if } (g \cdot x)|_F \in L \text{ for all } g \in G.$$

Definition 1.29 (Nearest neighbor SFT). Suppose G is generated by a finite symmetric set S . An SFT $X \subseteq \Sigma^G$ is *nearest neighbor* if it can be defined by a set of forbidden patterns all of whose supports are contained in $\{1_G, s\}$ for some $s \in S$. Equivalently, X is given by a family $(\mathcal{P}_s)_{s \in S}$ of subsets $\mathcal{P}_s \subseteq \Sigma \times \Sigma$ of allowed pairs, and

$$X = \{x \in \Sigma^G : (x_g, x_{gs}) \in \mathcal{P}_s \text{ for all } g \in G, s \in S\}.$$

Wang shifts are exactly the nearest neighbor SFTs on $\mathbb{Z}^2$, up to conjugacy.

Remark 1.30. Aperiodic sets of Wang tiles, that is, finite T with $X_T \neq \emptyset$ and no periodic configuration in X_T , were constructed by Berger [Ber66] and Robinson [Rob71], and much smaller ones by Kari [Kar96]. A minimal example (in terms of the number of Wang tiles) was found by Jeandel and Rao [JR21].

Remark 1.31. Every SFT on a finitely generated group is topologically conjugate to a nearest neighbor SFT, by passing to an alphabet whose letters are the allowed patterns with support a large ball. This elementary proof can be found in [Bar17], although we suggest it as an exercise to the reader.

Definition 1.32 (Sofic subshift). A subshift $X \subseteq \Sigma^G$ is *sofic* if it is the topological factor of an SFT on G .

Remark 1.33. Sofic subshifts were introduced by Weiss [Wei73]. One may equivalently define them as images of SFTs under sliding block codes. By Theorem 1.25 the first two definitions agree. Alternatively, we may define sofic subshifts as images of nearest neighbor SFTs under 1-block maps. This follows from a recoding as in Remark 1.31.

It is clear from the definition that every SFT on G is sofic (as it is the image of itself under the identity map). We will show that on every countable group which is not locally finite there exists a sofic subshift which is not an SFT.

Example 1.34 (The even shift). Consider the alphabet $\Sigma = \{0, 1\}$ and let $S \subseteq G$ be a finite symmetric subset which generates an infinite subgroup of G . Consider the Cayley graph $\Gamma = \text{Cay}(G, S)$ and let

$$X_{\text{even}} = \{x \in \{0, 1\}^G : \text{every finite connected component of 1s in } \Gamma \text{ has even size}\}.$$

It is easy to prove that X_{even} is not an SFT (think about what the language of X_{even} looks like and recall that we may assume that all forbidden patterns have a fixed finite support). On the other hand, we may consider the SFT Y on the alphabet $\Sigma' = \{T \subset S : |T| \text{ is odd or } T = \emptyset\}$ and given by the nearest neighbor rule: $y \in Y$ if and only if for every $g \in G$ if $s \in y(g)$ then $s^{-1} \in y(gs)$. Consider the 1-block map $\phi: Y \rightarrow X_{\text{even}}$ that sends the symbol $\emptyset$ to 0 and every other symbol to 1.

Let us check that $\phi(Y) \subset X_{\text{even}}$. Given $y \in Y$, consider the graph Γ_y whose vertices are the $g \in G$ with $y(g) \neq \emptyset$ and whose edges are the pairs $\{g, gs\}$ with $s \in y(g)$; the nearest neighbor rule is exactly what makes this set of edges well defined. By construction the degree of a vertex g in Γ_y is $|y(g)|$, which is odd, hence by the handshaking lemma every finite connected component of Γ_y has an even number of vertices. Now every edge of Γ_y is an edge of Γ joining two vertices which carry the symbol 1 in $\phi(y)$, hence every finite connected component of 1s in Γ is a disjoint union of connected components of Γ_y , and thus its size is a sum of even numbers. We leave the proof of surjectivity as an exercise.

The next class of interest for us are effective subshifts, those that can be defined by a recursively enumerable set of forbidden patterns. To that end, we shall give a brief introduction to computability theory.

1.5 Computability theory 101

We recall the notions from computability theory that we shall use. Everything in this subsection is classical; we only fix terminology and refer the reader to Rogers [Rog67] or Soare [Soa87] for complete treatments, and to [Tur36] for the original source.

Definition 1.35 (Turing machine). A *Turing machine* is a tuple $M = (Q, \Gamma, \sqcup, q_0, q_{\text{halt}}, \delta)$ where Q is a finite set of *states*, Γ a finite *tape alphabet* containing a distinguished *blank* symbol $\sqcup$, $q_0 \in Q$ the initial state, $q_{\text{halt}} \in Q$ a halting state, and

$$\delta: \Gamma \times (Q \setminus \{q_{\text{halt}}\}) \longrightarrow \Gamma \times Q \times \{-1, 0, 1\}$$

the *transition function*. The transition function is undefined on the halting state, which is what makes the machine stop.

We want to think of $\mathbb{Z}$ as a bi-infinite tape filled with symbols from Γ , accompanied by a head which carries a state and points to a position of $\mathbb{Z}$. This whole idea can be represented by an element of $\Gamma^{\mathbb{Z}} \times Q \times \mathbb{Z}$. In this sense, a Turing machine M acts on $\Gamma^{\mathbb{Z}} \times Q \times \mathbb{Z}$ as follows: let $(x, q, n) \in \Gamma^{\mathbb{Z}} \times Q \times \mathbb{Z}$ with $q \neq q_{\text{halt}}$ and let $\delta(x_n, q) = (a, r, d)$. Then $M(x, q, n) = (\tilde{x}, r, n + d)$ where $\tilde{x}_n = a$ and $\tilde{x}|_{\mathbb{Z} \setminus \{n\}} = x|_{\mathbb{Z} \setminus \{n\}}$. See Figure 2.

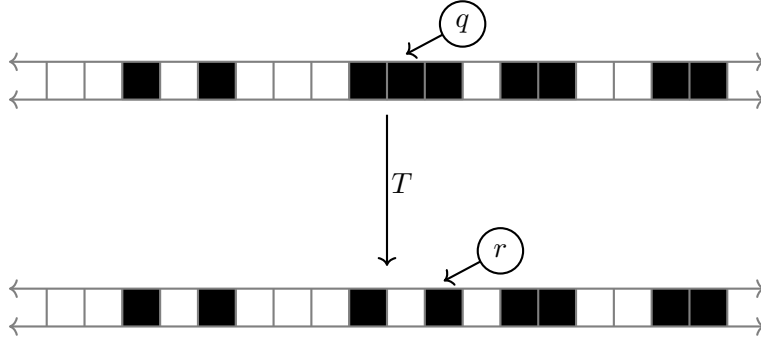


Figure 2: A Turing machine transition where $\delta(\blacksquare, q) = (\square, r, +1)$.

Let $\Sigma \subset \Gamma \setminus \{\sqcup\}$. A word $w \in \Sigma^*$ is called in this context an *input*. We say that a Turing machine M *halts* on an input w if the resulting sequence of configurations starting on a tape with w at the origin and blank symbols everywhere else eventually reaches q_{halt} .

Definition 1.36 (Computable and recursively enumerable sets). A language $L \subseteq \Sigma^*$ is

1. *recursively enumerable* if there is a Turing machine which halts on input w if and only if $w \in L$.
2. *decidable* if both L and its complement are recursively enumerable.

The notion of computation for Turing machines is a way to properly formalize the notion of algorithm. It is a formative exercise to represent the standard programming commands (such as FOR and WHILE loops) using Turing machine gadgets. This can be formalized a bit more precisely using the language of recursive functions.

Remark 1.37. It is direct from the definition that up to isomorphism there are countably many Turing machines, hence it follows that there exist languages which are not recursively enumerable. Some of these, such as the complement of the *halting problem*, can be constructed explicitly.

Turing machines also induce a notion of computability between words.

Definition 1.38. A function $f: A \subset \Sigma^* \rightarrow \Sigma'^*$ is called *computable* if there exists a Turing machine which on input $w \in A$ halts containing $f(w)$ on the tape. In the case when $A = \Sigma^*$ we say that f is *total computable*.

Remark 1.39 (Computability on countable sets). All the notions above are stated for subsets of Σ^* , but we shall apply them to subsets of other countable sets, such as the set of non-negative integers $\mathbb{N}$. To do so one fixes once and for all an explicit *encoding*. For instance, we may encode the non-negative integers as words in $\{0, 1\}^*$ through their binary expansion.

1.6 Effectively closed subshifts and the word problem

From now on G is a finitely generated group, and we fix a finite symmetric generating set $S = S^{-1}$. We denote by S^* the set of finite words over S , including the empty word ε , and by $w \mapsto \underline{w} \in G$ the evaluation map sending a word to the group element it represents. The map $w \mapsto \underline{w}$ is surjective by the choice of S .

Definition 1.40 (Word problem). The *word problem* of G with respect to S is the set

$$\text{WP}_S(G) = \{ w \in S^* : \underline{w} = 1_G \} \subseteq S^*.$$

We say that G is *recursively presented* if $\text{WP}_S(G)$ is recursively enumerable, and that it has *decidable word problem* if $\text{WP}_S(G)$ is a decidable subset of S^* .

Proposition 1.41. *Decidability of the word problem does not depend on the choice of the finite generating set S .*

Proof sketch. If S' is another finite generating set, each element of S' can be written as a fixed word in S and conversely; substituting these words gives a computable map $S'^* \rightarrow S^*$ preserving the represented group element, hence one may use the algorithm for $\text{WP}_S(G)$ to enumerate or decide $\text{WP}_{S'}(G)$. $\square$

It follows that the property of having decidable word problem (or being recursively presented) does not depend upon the particular choice of generating set, hence the notions are well-defined.

Remark 1.42. Free groups, finitely generated abelian groups, finitely generated nilpotent groups, hyperbolic groups and automatic groups all have decidable word problem. By the Novikov–Boone theorem [Nov55, Boo59] there exist finitely presented groups whose word problem is undecidable, and by the Higman embedding theorem [Hig61] every finitely generated and recursively presented group embeds into a finitely presented one; see [LS77] for beautiful proofs of both statements.

Next we shall define an intrinsic notion of computability for maps on subsets of the Cantor space.

Definition 1.43. A set $X \subset \Sigma^{\mathbb{N}}$ is called *effectively closed* if there exists a recursively enumerable language $L \subset \Sigma^*$ such that

$$X = \Sigma^{\mathbb{N}} \setminus \bigcup_{w \in L} [w],$$

where $[w]$ is the set of all infinite words that begin by w .

Definition 1.44. A map $f: X \subset \Sigma^{\mathbb{N}} \rightarrow \Sigma^{\mathbb{N}}$ is called *computable* if there exists a partial computable function $g: \Sigma^* \rightarrow \Sigma^*$ such that for any $x \in X$, $g(x|_{\{0, \dots, n\}})$ is defined for every n , the cylinders $[g(x|_{\{0, \dots, n\}})]$ form a nested sequence and

$$\{f(x)\} = \bigcap_{n \in \mathbb{N}} [g(x|_{\{0, \dots, n\}})].$$

In other words, f is computable if there is an algorithm which, given a prefix of $x \in X$, produces an approximation of $f(x)$, in such a way that $f(x)$ is approximated to an arbitrary precision. It is easy to show that computable maps are continuous (for instance, see [BCVR24]). In all statements below we assume that G is finitely generated.

Definition 1.45. An action $G \curvearrowright X \subset \Sigma^{\mathbb{N}}$ is called *computable* if X is an effectively closed set and each generator acts by computable maps.

Definition 1.46. A subshift $X \subset \Sigma^G$ is called *effective* (or Π_1^0) if it is topologically conjugate to a computable action $G \curvearrowright X \subseteq \{0, 1\}^{\mathbb{N}}$.

Remark 1.47. Whenever G is infinite there are uncountably many subshifts of $\{0, 1\}^G$ up to topological conjugacy. On the other hand, there are only countably many recursively enumerable languages, and thus only countably many effective subshifts. Hence most subshifts are not Π_1^0 , even though exhibiting a concrete one requires choosing a concrete non-recursively enumerable set.

It is sometimes useful to have an alternative definition of effectively closed subshifts directly through patterns. This can be done for recursively presented groups in a natural manner. See [ABS17] for further details.

Definition 1.48 (Pattern codings). A *pattern coding* over (S, Σ) is a finite set

$$c = \{(w_1, a_1), \dots, (w_k, a_k)\} \subseteq S^* \times \Sigma.$$

A configuration $x \in \Sigma^G$ *realizes* c at $g \in G$ if $x_{gw_i} = a_i$ for every $i \leq k$, and c *occurs* in x if it is realized at some $g \in G$. Given a set $\mathcal{C}$ of pattern codings we let, as before,

$$X_{\mathcal{C}} = \{x \in \Sigma^G : \text{no } c \in \mathcal{C} \text{ occurs in } x\}.$$

Remark 1.49. A pattern coding need not define a pattern: if $w_i = w_j$ while $a_i \neq a_j$, then no configuration realizes c anywhere and c is vacuous. Conversely each pattern admits infinitely many codings, and deciding whether two codings describe the same pattern amounts to solving the word problem. The set $S^* \times \Sigma$ is recursively enumerable regardless of G , which is exactly why this encoding is useful when generalizing beyond recursively presented groups.

Definition 1.50. A subshift $X \subseteq \Sigma^G$ is *effectively closed by patterns*, if $X = X_{\mathcal{C}}$ for some recursively enumerable set $\mathcal{C}$ of pattern codings.

Remark 1.51. Definition 1.50 does not depend on the choice of the finite generating set S , and if G is recursively presented it agrees with the notion of effective subshift. In non recursively presented groups the notion of effectively closed by patterns is strictly weaker.

Proposition 1.52. Let G be recursively presented, then every sofic subshift of Σ^G is effective.

Proof sketch. Let $X = \phi(Y)$, where $Y = Y_{\mathcal{F}} \subseteq \Sigma'^G$ is an SFT and ϕ is a 1-block code, as we may assume by the remark above. A pattern q fails to be globally admissible for X precisely when, for some finite $F \subseteq G$ containing $\text{supp}(q)$, no pattern of Σ'^F that is locally admissible for $\mathcal{F}$ projects to q on $\text{supp}(q)$; this is a consequence of compactness. As G is recursively presented, one can approximate the finite subsets of G to an arbitrary precision and check local admissibility, so the complement of $\mathcal{L}(X)$ is recursively enumerable. $\square$

Remark 1.53. The hypothesis in Proposition 1.52 cannot simply be dropped: the argument above uses in an essential way that one can effectively manipulate the finite portions of G . The relation between the word problem of G and the effectiveness of its sofic subshifts is subtle; see [BCVR24].

From what we have seen in this section, for every recursively presented group G , we have

$$\{\text{SFTs}\} \subseteq \{\text{sofic}\} \subseteq \{\text{effective}\}.$$

We have seen that the first inclusion is strict whenever G is not locally finite. The strictness of the second inclusion is the guiding topic for the next two lectures.

2 Lecture II: Self-simulation

The name simulation refers to the phenomenon where a complicated object (such as a computable action) can be encoded by a simpler object (such as an SFT). In this lecture we will first give an overview of the first simulation results in the context of dynamical systems in the literature, and then focus on a particular case of interest within the family of nonamenable groups.

In this section we shall assume implicitly that every group G we consider is infinite, finitely generated and recursively presented.

2.1 Simulation theorems

The first simulation theorem is due to Hochman, and shows that computable actions of $\mathbb{Z}$ can be represented through $\mathbb{Z}^3$ -SFTs.

Theorem 2.1 (Hochman [Hoc09]). *For every computable action $\mathbb{Z} \curvearrowright \{0, 1\}^{\mathbb{N}}$ there exists a $\mathbb{Z}^3$ -SFT whose $\mathbb{Z}$ -subaction is an extension of $\mathbb{Z} \curvearrowright \{0, 1\}^{\mathbb{N}}$.*

This result was the starting point for understanding subshifts of finite type in higher-dimensional groups as objects of a recursive nature. It led to the classification of topological entropies of $\mathbb{Z}^2$ -SFTs by Hochman and Meyerovitch [HM10].

This theorem was later improved for the expansive case independently by Aubrun and Sablik [AS13] and by Durand, Romashchenko and Shen [DRS12] who proved the following statement.

Theorem 2.2 ([AS13, DRS12]). *For every effective subshift $X \subset \Sigma^{\mathbb{Z}}$ its natural lift to $\mathbb{Z}^2$ is sofic.*

Let us recall that if $\phi: G \rightarrow H$ is a group epimorphism and $X \subset \Sigma^H$ is a subshift, one can define its lift to G as the subshift $\phi^*(X) \subset \Sigma^G$ where

$$\phi^*(X) = \left\{ \left(x_{\phi(g)} \right)_{g \in G} : x \in X \right\}.$$

In other words, one can picture the lift of $X \subset \Sigma^{\mathbb{Z}}$ as the $\mathbb{Z}^2$ -subshift whose columns are constant and whose rows carry a configuration of X .

For general groups, we may give the following notion of simulation.

Definition 2.3. Let $\phi: G \rightarrow H$ be a group epimorphism. We say that G simulates H if the lift to G of every effective H -subshift is sofic.

Theorem 2.4. *We have the following simulation results:*

1. [BS19] *For every finitely generated H and $d \geq 2$, we have that $G = \mathbb{Z}^d \rtimes H$ simulates H .*
2. [Bar19] *For finitely generated H, N_1, N_2 , the group $G = H \times N_1 \times N_2$ simulates H .*

In the two results above, the epimorphisms are the canonical ones.

A common point on these simulation results is that in all of them the groups H are simulated by “larger” groups G . This leads to the natural question: is there a group which simulates itself through the identity map? Equivalently, is there a group where every effective subshift is sofic? This leads to the notion of self-simulable groups.

Definition 2.5. A group G is called self-simulable if every effective G -subshift is sofic.

We note that the original definition in [BSS26] is different, it asks instead that every computable action is the topological factor of an SFT. It turns out that these notions are equivalent (for recursively presented groups) for two reasons that shall be justified later: first, no amenable group can be self-simulable, and second, every computable action of a non-amenable group is the topological factor of an effective G -subshift.

2.2 Amenability

Amenability is a fundamental notion needed to state the results of this lecture: no amenable group is self-simulable, while many non-amenable groups are. In this subsection we introduce amenability through Følner sequences and give the combinatorial characterizations of non-amenability that will be used later. A thorough treatment can be found in [CSC10, Chapter 4].

Definition 2.6 (Følner sequence). Let G be a countable group. A sequence $(F_n)_{n \in \mathbb{N}}$ of nonempty finite subsets of G is a *Følner sequence* if for every $g \in G$ we have

$$\lim_{n \rightarrow \infty} \frac{|gF_n \Delta F_n|}{|F_n|} = 0.$$

The group G is *amenable* if it admits a Følner sequence.

In other words, G is amenable if it contains finite subsets which are almost invariant under translation.

Remark 2.7. If (F_n) is a Følner sequence then so is (F_n^{-1}) for right translations, since $|F_n^{-1}g^{-1} \Delta F_n^{-1}| = |(gF_n)^{-1} \Delta F_n^{-1}| = |gF_n \Delta F_n|$. We call (F_n^{-1}) a *right Følner sequence* and amenability can obviously be also defined using these sets instead.

When G is generated by a finite set S , it suffices to check the Følner condition for $g \in S$, because $|ghF \Delta F| \leq |ghF \Delta gF| + |gF \Delta F| = |hF \Delta F| + |gF \Delta F|$.

Example 2.8 (Finite groups). If G is finite, then $F_n = G$ defines a Følner sequence, thus every finite group is amenable.

Example 2.9 ($\mathbb{Z}$ and $\mathbb{Z}^d$). In $\mathbb{Z}$ take $F_n = \{0, 1, \dots, n\}$. For $g \in \mathbb{Z}$ the symmetric difference $|(g + F_n) \Delta F_n| \leq 2|g|$ is bounded, while $|F_n| = n + 1$ tends to infinity, so the quotient tends to 0. In $\mathbb{Z}^d$ the boxes $F_n = \{0, \dots, n\}^d$ work in the same way: the symmetric difference with a translate is contained in a bounded neighborhood of the boundary of the box, which has $O(n^{d-1})$ elements, while $|F_n| = (n + 1)^d$.

The two previous examples are instances of a general principle: a group cannot be non-amenable unless its balls grow exponentially.

Proposition 2.10. Let G be generated by a finite symmetric set S with $1_G \in S$ and let $B_n = S^n$ denote the ball of radius n . If G has subexponential growth, that is, if $\lim_{n \rightarrow \infty} |B_n|^{1/n} = 1$, then G is amenable.

Proof. Fix $\varepsilon > 0$ and suppose, towards a contradiction, that $|B_{n+1}| > (1 + \varepsilon)|B_n|$ for every $n \in \mathbb{N}$. Iterating, $|B_n| \geq (1 + \varepsilon)^n |B_0| = (1 + \varepsilon)^n$, so $\liminf_n |B_n|^{1/n} \geq 1 + \varepsilon$, contradicting subexponential growth. Hence for every $\varepsilon > 0$ there are infinitely many n with $|B_{n+1}| \leq (1 + \varepsilon)|B_n|$. Choosing for each $k \geq 1$ such an n_k with $\varepsilon = 1/k$, we obtain finite sets $F_k = B_{n_k}$ which satisfy, for every $s \in S$,

$$|sF_k \Delta F_k| = 2|sF_k \setminus F_k| \leq 2|B_{n_k+1} \setminus B_{n_k}| \leq \frac{2}{k}|F_k|,$$

where we used $sB_{n_k} \subseteq B_{n_k+1}$ and $|sF_k| = |F_k|$. By Remark 2.7 it suffices to check the Følner condition on S , so $(F_k)_k$ is a Følner sequence. $\square$

Example 2.11 (Abelian and nilpotent groups). Finitely generated abelian groups have polynomial growth, and so do finitely generated nilpotent groups by the Bass–Guivarc’h theorem. By Proposition 2.10 they are all amenable. More generally, amenability is preserved under taking subgroups, quotients, extensions and directed unions, which yields the amenability of all solvable groups and of all locally finite groups; see [CSC10, Chapter 4].

It will be convenient to restate the Følner condition in a form which involves a single inequality. Write $FK = \{fk : f \in F, k \in K\}$.

Lemma 2.12. *Let G be a countable group. The following are equivalent.*

1. G is amenable.
2. For every $K \subseteq G$ and every $\varepsilon > 0$ there exists a nonempty $F \subseteq G$ such that $|FK| \leq (1 + \varepsilon)|F|$.

Consequently, G is non-amenable if and only if there are $K \subseteq G$ with $1_G \in K$ and $\varepsilon > 0$ such that $|FK| \geq (1 + \varepsilon)|F|$ for every nonempty $F \subseteq G$.

Proof. Suppose G is amenable and let (F_n) be a right Følner sequence as in [Remark 2.7](#). Given $K \subseteq G$ and $\varepsilon > 0$, choose n large enough that $|F_n k \Delta F_n| \leq \frac{\varepsilon}{|K|}|F_n|$ for every $k \in K$. As $FK = \bigcup_{k \in K} Fk$, we get

$$|F_n K| \leq |F_n| + \sum_{k \in K} |F_n k \setminus F_n| \leq (1 + \varepsilon)|F_n|.$$

Conversely, assume (2) and let $K \subseteq G$; we may assume $1_G \in K$ and $K = K^{-1}$. Given $\varepsilon > 0$, pick a nonempty $F \subseteq G$ with $|FK| \leq (1 + \varepsilon)|F|$. For $k \in K$ we have $Fk \subseteq FK$, so $|Fk \setminus F| \leq |FK \setminus F| \leq \varepsilon|F|$, and since $|Fk| = |F|$ we also get $|F \setminus Fk| = |Fk \setminus F|$. Hence $|Fk \Delta F| \leq 2\varepsilon|F|$ for every $k \in K$. Taking an exhaustion of G by finite sets K_n and $\varepsilon = 1/n$ produces a right Følner sequence, and inverting its terms a Følner sequence.

The last assertion is the negation of (2), together with the observation that enlarging K only makes $|FK|$ larger, so we may always assume $1_G \in K$. $\square$

Paradoxical decompositions

Amenability admits a purely combinatorial opposite: a group is non-amenable exactly when it can be cut into finitely many pieces which can be rearranged, by translations, into two copies of the group.

Definition 2.13 (Paradoxical decomposition). A group G is *paradoxical* if there exist pairwise disjoint subsets $A_1, \dots, A_m, B_1, \dots, B_n \subseteq G$ and elements $g_1, \dots, g_m, h_1, \dots, h_n \in G$ such that

$$G = \bigsqcup_{i=1}^m A_i g_i = \bigsqcup_{j=1}^n B_j h_j.$$

Remark 2.14. We have written the definition with right translations because this is the form we shall use. It is equivalent to the more common formulation with left translations: the map $g \mapsto g^{-1}$ sends a decomposition as above to the sets A_i^{-1}, B_j^{-1} and the elements g_i^{-1}, h_j^{-1} , and $(A_i g_i)^{-1} = g_i^{-1} A_i^{-1}$.

Example 2.15 (The free group). Let $\mathbb{F}_2 = \langle a, b \rangle$ be the free group of rank 2. For a generator $x \in \{a^{\pm 1}, b^{\pm 1}\}$ let $W(x) \subseteq \mathbb{F}_2$ be the set of nontrivial reduced words which end with the letter x . As every nontrivial reduced word ends with exactly one letter, the four sets $W(a), W(a^{-1}), W(b), W(b^{-1})$ are pairwise disjoint.

We claim that $\mathbb{F}_2 = W(a) \sqcup W(a^{-1})a$. Indeed, let $u \in \mathbb{F}_2$. If u ends with a , then $u \in W(a)$, and $u \notin W(a^{-1})a$: writing $u = va$ with v reduced and not ending with a^{-1} , we have $ua^{-1} = v \notin W(a^{-1})$. If u does not end with a , which includes the case $u = 1$, then ua^{-1} is reduced and ends with a^{-1} , so $u = (ua^{-1})a \in W(a^{-1})a$. The same argument gives $\mathbb{F}_2 = W(b) \sqcup W(b^{-1})b$.

Thus $\mathbb{F}_2$ is paradoxical, with $A_1 = W(a)$, $A_2 = W(a^{-1})$, $B_1 = W(b)$, $B_2 = W(b^{-1})$ and $g_1 = h_1 = 1$, $g_2 = a$, $h_2 = b$. Note that the four pieces do not cover $\mathbb{F}_2$, as the identity belongs to none of them; this is why [Definition 2.13](#) only requires the pieces to be pairwise disjoint.

Remark 2.16. Any group G containing a copy of $\mathbb{F}_2$ is paradoxical. Indeed, let $T \subseteq G$ be a set of representatives of the left cosets of $\mathbb{F}_2$, so that $G = \bigsqcup_{t \in T} t\mathbb{F}_2$, and let A_i, B_j, g_i, h_j be as in [Example 2.15](#). Putting $\widetilde{A}_i = \bigsqcup_{t \in T} tA_i$ and $\widetilde{B}_j = \bigsqcup_{t \in T} tB_j$, we obtain $\bigsqcup_i \widetilde{A}_i g_i = \bigsqcup_{t \in T} t(\bigsqcup_i A_i g_i) = \bigsqcup_{t \in T} t\mathbb{F}_2 = G$, and similarly for the $\widetilde{B}_j$.

The link between paradoxicality and the existence of the 2-to-1 maps we shall use later is provided by the following classical result, a version of Hall's marriage theorem for locally finite bipartite graphs. Recall that for a set S of vertices, $N(S)$ denotes the set of vertices adjacent to some element of S .

Theorem 2.17 (Hall's 2-to-1 theorem). *Let Γ be a bipartite graph with parts X and Y in which every vertex has finite degree. There exists a map $\psi: Y \rightarrow X$ such that $\psi(y)$ is adjacent to y for every $y \in Y$ and $|\psi^{-1}(x)| = 2$ for every $x \in X$, if and only if*

$$|N(S)| \geq 2|S| \text{ for every finite } S \subseteq X, \quad |N(T)| \geq \frac{1}{2}|T| \text{ for every finite } T \subseteq Y.$$

We are ready for the characterization.

Theorem 2.18. *Let G be a countable group. The following are equivalent.*

1. G is non-amenable.
2. G is paradoxical.
3. There exist $K \in G$ and a 2-to-1 map $\varphi: G \rightarrow G$ such that $g^{-1}\varphi(g) \in K$ for every $g \in G$.

Here 2-to-1 means that $|\varphi^{-1}(h)| = 2$ for every $h \in G$. The condition $g^{-1}\varphi(g) \in K$ says that φ moves every element of G by a bounded amount, namely $\varphi(g) \in gK$.

Proof. (1) $\Rightarrow$ (3). Suppose G is non-amenable. By [Lemma 2.12](#) there are $K_0 \in G$ with $1_G \in K_0$ and $\varepsilon > 0$ such that $|FK_0| \geq (1 + \varepsilon)|F|$ for every nonempty $F \in G$. Applying this repeatedly, $|FK_0^n| \geq (1 + \varepsilon)^n|F|$, so choosing n with $(1 + \varepsilon)^n \geq 2$ and setting $K = (K_0^n)^{-1}$ we obtain $1_G \in K$ and

$$|SK^{-1}| \geq 2|S| \quad \text{for every nonempty } S \in G.$$

Consider now the bipartite graph Γ whose two parts X and Y are copies of G , and in which $y \in Y$ is adjacent to $x \in X$ if and only if $x \in yK$. Every vertex has degree at most $|K|$, so Γ is locally finite. For a finite $S \subseteq X$ we have $N(S) = \{y : yK \cap S \neq \emptyset\} = SK^{-1}$, so $|N(S)| \geq 2|S|$; and for a finite $T \subseteq Y$ we have $N(T) = TK \supseteq T$ because $1_G \in K$, so $|N(T)| \geq |T| \geq \frac{1}{2}|T|$. By [Theorem 2.17](#) there is a map $\varphi: Y \rightarrow X$, that is, a map $\varphi: G \rightarrow G$, such that $\varphi(g) \in gK$ for every g and $|\varphi^{-1}(h)| = 2$ for every h . This is precisely (3).

(3) $\Rightarrow$ (2). Let φ and $K = \{k_1, \dots, k_m\}$ be as in (3). For every $h \in G$ the fiber $\varphi^{-1}(h)$ has exactly two elements, so we may choose maps $\alpha, \beta: G \rightarrow G$ with $\varphi^{-1}(h) = \{\alpha(h), \beta(h)\}$ and $\alpha(h) \neq \beta(h)$. Put $A = \alpha(G)$ and $B = \beta(G)$. Since the fibers of φ partition G , the sets A and B are disjoint and $A \sqcup B = G$; moreover α and β are injective, because $h = \varphi(\alpha(h))$, so the restrictions $\varphi|_A: A \rightarrow G$ and $\varphi|_B: B \rightarrow G$ are bijections.

Split A according to which element of K realizes the displacement: let

$$A_i = \{g \in A : i \text{ is minimal with } \varphi(g) = gk_i\}, \quad 1 \leq i \leq m,$$

and define $B_i \subseteq B$ in the same way. The sets $A_1, \dots, A_m$ partition A , the sets $B_1, \dots, B_m$ partition B , and all of them are pairwise disjoint. By construction $\varphi(A_i) = A_i k_i$, and as $\varphi|_A$ is

a bijection onto G whose restriction to each A_i is injective, the sets $A_i k_i$ are pairwise disjoint with union G :

$$G = \varphi(A) = \bigsqcup_{i=1}^m A_i k_i, \quad \text{and likewise} \quad G = \varphi(B) = \bigsqcup_{i=1}^m B_i k_i.$$

Hence G is paradoxical, with $g_i = h_i = k_i$.

(2) $\Rightarrow$ (1). Suppose G is paradoxical, with pieces A_i, B_j and elements g_i, h_j as in [Definition 2.13](#), and suppose towards a contradiction that G is amenable. Let

$$K = \{g_1^{-1}, \dots, g_m^{-1}, h_1^{-1}, \dots, h_n^{-1}\}$$

and let $F \subseteq G$ be nonempty with $|Fk \setminus F| \leq \varepsilon|F|$ for every $k \in K$, where $\varepsilon = \frac{1}{2(m+n)}$; such an F exists by [Remark 2.7](#) and [Lemma 2.12](#). Since $A_i g_i \cap F$ is in bijection with $A_i \cap F g_i^{-1}$ we get

$$|A_i g_i \cap F| = |A_i \cap F g_i^{-1}| \leq |A_i \cap F| + |F g_i^{-1} \setminus F| \leq |A_i \cap F| + \varepsilon|F|,$$

and the same bound holds for the B_j . Summing over i and using that the sets $A_i g_i$ partition G , and then doing the same for the B_j , we obtain

$$2|F| = \sum_{i=1}^m |A_i g_i \cap F| + \sum_{j=1}^n |B_j h_j \cap F| \leq \sum_{i=1}^m |A_i \cap F| + \sum_{j=1}^n |B_j \cap F| + (m+n)\varepsilon|F|.$$

The sets A_i, B_j are pairwise disjoint, so the first two sums add up to at most $|F|$. Therefore $2|F| \leq |F| + \frac{1}{2}|F|$, which is absurd. Hence G is non-amenable. $\square$

Corollary 2.19. *The free group $\mathbb{F}_2$ is non-amenable, and so is every group containing a copy of $\mathbb{F}_2$.*

Remark 2.20. The converse of the corollary is false: there exist finitely generated non-amenable groups with no free subgroup of rank 2, the first examples being Ol'shanskii's Tarski monsters. This was the celebrated von Neumann problem.

2.3 The paradoxical subshift

The characterization of non-amenableity given by [Theorem 2.18](#) is local in nature: the map φ moves every element of G by an amount taken from a fixed finite set K , and being 2-to-1 can be verified by looking, around each element, at a window of size $|K|$. Both features together say that these maps are the configurations of an SFT. This is the *paradoxical subshift* of [\[BSS26\]](#), and the reason for which we shall encode not just φ but also a coloring is that we want to extract from each configuration a family of *disjoint* one-sided infinite paths, indexed by the elements of the group.

Throughout this subsection $K \subseteq G$ is a symmetric set which contains 1_G . We use the two colors $\mathbf{G}$ (green) and $\mathbf{B}$ (blue), and write $\bar{t}$ for the color opposite to t .

Definition 2.21 (The alphabet). Let $\Sigma_K = K^3 \times \{\mathbf{G}, \mathbf{B}\}$. Given a symbol $a = ((a_1, a_2, a_3), t) \in \Sigma_K$ we write

$$L_{\mathbf{G}}(a) = a_1, \quad L_{\mathbf{B}}(a) = a_2, \quad R_t(a) = a_3, \quad t(a) = t,$$

and call $t(a)$ the *color* of a . Note that R is only defined for the color of the symbol which carries it. The letters L and R stand for left and right.

A symbol thus carries three displacements: two *incoming* arrows, one of each color, and one *outgoing* arrow, whose color is the color of the symbol. The subshift is defined by requiring that these arrows match.

Definition 2.22 (Paradoxical subshift). The *paradoxical subshift* is the G -subshift $P \subseteq \Sigma_K^G$ consisting of all ρ such that for every $g \in G$, writing $a = \rho(g)$:

1. $b = \rho(gL_G(a))$ has color $\mathbf{G}$ and $R_G(b) = (L_G(a))^{-1}$,
2. $c = \rho(gL_B(a))$ has color $\mathbf{B}$ and $R_B(c) = (L_B(a))^{-1}$,
3. if $d = \rho(gR_{t(a)}(a))$, then $L_{t(a)}(d) = (R_{t(a)}(a))^{-1}$.

The three conditions say that following an arrow of a given color and then following back the arrow of the same color amounts to no movement at all. A configuration of P therefore describes a directed graph on G in which every vertex has exactly one outgoing arrow and exactly two incoming arrows, one green and one blue. A picture of the local structure may be found in Figure 3.

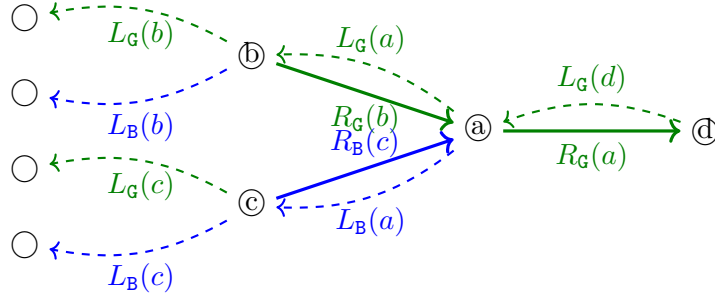


Figure 3: The local structure of the paradoxical subshift.

Proposition 2.23. P is a G -subshift of finite type.

Proof. Each of the three conditions at g involves only the coordinates of ρ in $g(K \cup \{1_G\})$, because all displacements are elements of K . Thus the set of patterns $p \in \Sigma_K^{K \cup \{1_G\}}$ for which one of the three conditions fails at 1_G is a finite set of forbidden patterns defining P . $\square$

Lemma 2.24. Let $\rho \in P$ and define $\varphi_\rho(g) = gR_{t(\rho(g))}(\rho(g))$. Then $\varphi_\rho: G \rightarrow G$ is a surjective 2-to-1 map with $g^{-1}\varphi_\rho(g) \in K$ for every $g \in G$. In particular, if $P \neq \emptyset$ then G is non-amenable.

Proof. The displacement $R_{t(\rho(g))}(\rho(g))$ lies in K by definition of the alphabet. Fix $h \in G$ and let $a = \rho(h)$. By condition (1) the element $u = hL_G(a)$ satisfies $R_G(\rho(u)) = (L_G(a))^{-1}$, hence $\varphi_\rho(u) = uR_G(\rho(u)) = hL_G(a)(L_G(a))^{-1} = h$, and u has color $\mathbf{G}$. By condition (2) the element $v = hL_B(a)$ likewise satisfies $\varphi_\rho(v) = h$ and has color $\mathbf{B}$; in particular $u \neq v$. Conversely, if $\varphi_\rho(w) = h$ and w has color t , then condition (3) applied to w gives $L_t(\rho(h)) = (R_t(\rho(w)))^{-1}$, so $hL_t(\rho(h)) = w$, that is, $w \in \{u, v\}$. Hence every fiber of φ_ρ has exactly two elements. The last claim follows from Theorem 2.18. $\square$

Proposition 2.25 ([BSS26, Proposition 4.3]). Let G be a non-amenable group. Then there is a symmetric $K \subseteq G$ containing 1_G for which the associated paradoxical subshift P is nonempty.

Proof. By Theorem 2.18 there are a finite set K and a surjective 2-to-1 map $\varphi: G \rightarrow G$ with $g^{-1}\varphi(g) \in K$ for every g . Replacing K by $K \cup K^{-1} \cup \{1_G\}$ we may assume that K is symmetric and contains the identity, so that $\varphi(g)^{-1}g \in K$ as well.

As φ is 2-to-1 and surjective, we may partition $G = G_G \sqcup G_B$ in such a way that the restrictions φ_G and φ_B of φ to G_G and G_B are bijections onto G : it suffices to choose, for each $h \in G$, which of the two preimages of h is green and which is blue. Define $\rho \in \Sigma_K^G$ by

$$\rho(g) = \left((g^{-1}\varphi_G^{-1}(g), g^{-1}\varphi_B^{-1}(g), g^{-1}\varphi(g)), t(g) \right),$$

where $t(g) = \mathbf{G}$ if $g \in G_{\mathbf{G}}$ and $t(g) = \mathbf{B}$ otherwise. The three displacements do lie in K : the third one by hypothesis, and for the first one, letting $h = \varphi_{\mathbf{G}}^{-1}(g)$ we have $g^{-1}\varphi_{\mathbf{G}}^{-1}(g) = \varphi(h)^{-1}h \in K$; the second one is identical. Conditions (1), (2) and (3) of [Definition 2.22](#) are now a direct verification. For instance, for (1), the element $u = gL_{\mathbf{G}}(\rho(g)) = \varphi_{\mathbf{G}}^{-1}(g)$ is green and satisfies $\varphi(u) = g$, so $R_{\mathbf{G}}(\rho(u)) = u^{-1}g = (L_{\mathbf{G}}(\rho(g)))^{-1}$. $\square$

Remark 2.26. P admits no shift-invariant Borel probability measure. Indeed, suppose μ were one and, for $k \in K$, let

$$A_k = \{ \rho \in P : R_{t(\rho(1_G))}(\rho(1_G)) = k \},$$

the set of configurations whose outgoing arrow at 1_G is k . These sets partition P , so $\sum_{k \in K} \mu(A_k) = 1$. On the other hand, $k^{-1} \cdot A_k = \{ \rho : R_{t(\rho(k^{-1}))}(\rho(k^{-1})) = k \}$, which is the set of configurations in which k^{-1} sends its arrow to 1_G , so by invariance

$$1 = \sum_{k \in K} \mu(A_k) = \sum_{k \in K} \mu(k^{-1} \cdot A_k) = \int_P |\varphi_{\rho}^{-1}(1_G)| d\mu(\rho) = 2,$$

where the middle equality holds because every preimage of 1_G is of the form k^{-1} with $k \in K$, and the last one by [Lemma 2.24](#). In particular no configuration of P has a stabilizer of finite index, and $P = \emptyset$ whenever G is amenable, since on an amenable group every nonempty subshift carries an invariant measure obtained by averaging along a Følner sequence.

We now extract the paths. Recall that $\bar{t}$ denotes the color opposite to t .

Definition 2.27 (The paths). For $g \in G$ define $\gamma_g : \mathbb{N} \times P \rightarrow G$ by

$$\gamma_g(n, \rho) = \begin{cases} gL_{\bar{t}}(\rho(g)) & \text{if } n = 0, \\ \gamma_g(n-1, \rho) L_{\bar{t}}(\rho(\gamma_g(n-1, \rho))) & \text{if } n > 0, \end{cases}$$

where $t = t(\rho(g))$ is the color of $\rho(g)$.

In words, γ_g assigns to g the path obtained by repeatedly following the incoming arrow of the color *opposite* to that of g . The map $\gamma_g(n, \cdot)$ is continuous, as $\gamma_g(n, \rho)$ depends only on the values of ρ on gK^{n+1} , and directly from the definitions we get the equivariance relation

$$\gamma_g(n, \rho) = g \gamma_{1_G}(n, g^{-1} \cdot \rho). \quad (1)$$

Remark 2.28. If $\rho(g)$ has color t , then $\rho(\gamma_g(n, \rho))$ has color $\bar{t}$ for every $n \in \mathbb{N}$, by conditions (1) and (2) of [Definition 2.22](#). Moreover, by condition (3), for every $n > 0$,

$$\gamma_g(n-1, \rho) = \gamma_g(n, \rho) R_{\bar{t}}(\rho(\gamma_g(n, \rho))) = \varphi_{\rho}(\gamma_g(n, \rho)).$$

That is, the path is a φ_{ρ} -orbit traversed backwards.

Lemma 2.29 ([BSS26, Lemma 4.5]). *For every $\rho \in P$ the map $(n, g) \mapsto \gamma_g(n, \rho)$ from $\mathbb{N} \times G$ to G is injective.*

Proof. Suppose $\gamma_g(n, \rho) = \gamma_h(m, \rho)$. By [Remark 2.28](#) the color of this common element is opposite both to that of $\rho(g)$ and to that of $\rho(h)$, so $\rho(g)$ and $\rho(h)$ have the same color, say t . Using the relation of [Remark 2.28](#) repeatedly, that is, applying φ_{ρ} to both sides $\min(n, m)$ times, we may assume that $n = 0$ or $m = 0$; without loss of generality $n = 0$.

Let $v = \gamma_g(0, \rho)$, so that $g = vR_{\bar{t}}(\rho(v))$ by [Remark 2.28](#) and $\rho(v)$ has color $\bar{t}$, different from that of $\rho(g)$. If we had $m > 0$, then $\rho(\gamma_h(m-1, \rho)) = \rho(\varphi_{\rho}(v)) = \rho(g)$ would have color $\bar{t}$, since every element of the path of h has the color opposite to $\rho(h)$, and $\rho(h)$ has color t . This contradicts the fact that $\rho(g)$ has color t . Hence $m = 0$, and then $h = vR_{\bar{t}}(\rho(v)) = g$. $\square$

So a configuration of P equips the group with a family $(\gamma_g)_{g \in G}$ of one-sided infinite paths with jumps in K , one for each element of the group, and these paths are pairwise disjoint and self-avoiding. This is the whole point of the construction: a path is a copy of $\mathbb{N}$, and it is placed inside the group by a rule which a sliding block code can read.

Remark 2.30. Families of this kind appeared before in geometric group theory, notably in Whyte's theorem [Why99] that every non-amenable group admits a translation-like action of $\mathbb{F}_2$. The subshift formulation is what allows one to impose, by local rules, conditions on what is written along the paths. As a first illustration, one obtains an alternative proof of a theorem of Seward [Sew14]: every action of a countable non-amenable group is a factor of a subshift, and if the action is computable the subshift can be taken effectively closed; see [BSS26, Theorem 4.6].

Finally, the reason for taking a direct product of two non-amenable groups.

Corollary 2.31. *Let $G = G_H \times G_V$ with G_H and G_V finitely generated and non-amenable. There are finite sets $K_H \subseteq G_H$, $K_V \subseteq G_V$ and a nonempty G -SFT $\tilde{P}$ such that every $\rho \in \tilde{P}$ induces maps $\gamma_g: \mathbb{N}^2 \rightarrow G$, $g \in G$, for which the map $(u, g) \mapsto \gamma_g(u, \rho)$ from $\mathbb{N}^2 \times G$ to G is injective, and such that $\gamma_g(u, \rho)$ depends on ρ only through finitely many coordinates.*

Proof. Let P_H and P_V be nonempty paradoxical subshifts on G_H and G_V , which exist by Proposition 2.25. Let $\tilde{P}_H$ be the G -subshift of configurations ρ on the alphabet of P_H which do not depend on the second coordinate and whose restriction to $G_H \times \{1\}$ lies in P_H , and define $\tilde{P}_V$ symmetrically. Both are G -SFTs, and so is $\tilde{P} = \tilde{P}_H \times \tilde{P}_V$. For $g = (g_1, g_2)$ and $\rho = (\rho_H, \rho_V)$ put

$$\gamma_g((n_1, n_2), \rho) = \left(\gamma_{g_1}^H(n_1, \rho_H), \gamma_{g_2}^V(n_2, \rho_V) \right),$$

where γ^H and γ^V are the maps of Definition 2.27 for each factor. Injectivity follows coordinatewise from Lemma 2.29. $\square$

Each $g \in G$ is thus the corner of its own copy of $\mathbb{N}^2$ inside G , and distinct elements get disjoint copies. In the next subsection we describe what we shall write on these grids.

2.4 Encoding computations in $\mathbb{N}^2$

The second ingredient is classical and goes back to Wang [Wan61] and Berger [Ber66]: the space-time diagram of a Turing machine is a tiling of $\mathbb{N}^2$ by a finite set of Wang tiles. We describe the tileset we need; details are in [BSS26, Section 5.1].

Let $M = (Q, \Gamma, \sqcup, q_0, q_{\text{halt}}, \delta)$ be a Turing machine working on a one-sided tape. Consider the finite set W_M of Wang tiles whose horizontal sides carry either nothing or a state of Q together with a direction, and whose vertical sides carry a symbol of Γ , possibly paired with a state. Among these tiles there are three distinguished ones, called **seed**, $\text{tile}_1(a)$ and $\text{tile}_2(a)$ for $a \in \Gamma$, which are used to write the initial tape, and one tile for each transition of δ . We let T_M be the set of tilings $\tau: \mathbb{N}^2 \rightarrow W_M$ in which adjacent tiles carry the same data on their common side, see Figure 4.

Remark 2.32. If $\tau(0, 0) = \text{seed}$, then necessarily $\tau(1, 0) = \text{tile}_1(a_1)$ and $\tau(n, 0) = \text{tile}_2(a_n)$ for every $n \geq 2$, and the tiles at all remaining positions of $\mathbb{N}^2$ are uniquely determined. The n -th row of τ is then the configuration of the tape after n steps of the computation of M on the input $(a_n)_{n \in \mathbb{N}}$. This is the only property of W_M that we shall use.

Let now $Y \subseteq \Sigma^{\mathbb{N}}$ be an effectively closed set and let M_Y be a Turing machine which halts on a word w if and only if $[w] \cap Y = \emptyset$. Let $M_{\text{search}, Y}$ be the machine which, on an input tape whose first component carries a sequence $(y_n)_{n \in \mathbb{N}}$, performs the following search: for $k = 1, 2, 3, \dots$, it runs M_Y for k steps on every word w of length at most k , and whenever M_Y

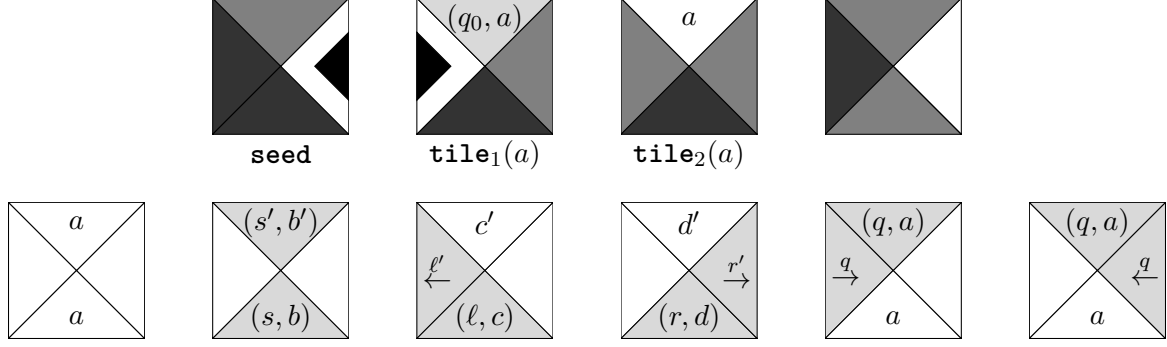


Figure 4: Wang tiles encoding the space-time diagram of a Turing machine.

halts on such a w it checks whether $y_0 \dots y_{|w|-1} = w$, halting if so. By construction $M_{\text{search}, Y}$ halts on $(y_n)_{n \in \mathbb{N}}$ if and only if $(y_n)_{n \in \mathbb{N}} \notin Y$.

Definition 2.33 (Search tilespace). $T_{\text{search}, Y}$ is the set of tilings $\tau \in T_{M_{\text{search}, Y}}$ such that $\tau(0, 0) = \text{seed}$, such that the tiles $\text{tile}_i(a, b)$ with $b \neq \sqcup$ are forbidden, and such that no tile carrying the final state q_{halt} occurs.

The first two conditions say that the bottom row of τ codes a sequence $(y_n)_{n \in \mathbb{N}}$ on a tape which is otherwise blank, and the third one says that the search never succeeds, that is, $(y_n)_{n \in \mathbb{N}} \in Y$.

Definition 2.34 (Synchronization tilespace). T_{sync} is the set of maps $\sigma: \mathbb{N}^2 \rightarrow \Sigma'$ such that $\sigma(u + (0, 1)) = \sigma(u + (1, 0))$ for every $u \in \mathbb{N}^2$, that is, the set of tilings whose antidiagonals are constant.

Definition 2.35 (Computation tilespace). $T_{\text{comp}, Y}$ is the set of pairs $(\tau, \sigma) \in T_{\text{search}, Y} \times T_{\text{sync}}$ such that for every $u \in \mathbb{N}^2$, if $\tau(u + (1, 0)) = \text{tile}_i(a, \sqcup)$ then $\sigma(u) = a$.

Proposition 2.36. *Let $Y \subseteq \Sigma'^{\mathbb{N}}$ be effectively closed. Then:*

1. *for every $(\tau, \sigma) \in T_{\text{comp}, Y}$ we have $(\sigma(0, n))_{n \in \mathbb{N}} = (\sigma(n, 0))_{n \in \mathbb{N}} \in Y$;*
2. *for every $y \in Y$ there is $(\tau, \sigma) \in T_{\text{comp}, Y}$ with $y = (\sigma(n, 0))_{n \in \mathbb{N}}$.*

Proof. For (1), the equality of the two sequences is the definition of T_{sync} , and the last condition ties the bottom row of σ to the input written on the bottom row of τ , which lies in Y because no tile carries q_{halt} . For (2), given $y \in Y$ let $\sigma(n - k, k) = y_n$ for all $k \leq n$, which is synchronized, and let τ be the unique tiling of Remark 2.32 with input y ; it has no occurrence of q_{halt} precisely because $y \in Y$. $\square$

The synchronization layer is what makes the two directions of the grid interchangeable: the sequence $y \in Y$ is written on the bottom row, on the left column, and on every antidiagonal, so it can be read while moving along either factor of G .

2.5 Products of non-amenable groups are self-simulable

Theorem 2.37 ([BSS26, Theorem 1.2]). *The direct product $G = G_H \times G_V$ of any two finitely generated non-amenable groups is self-simulable. That is, every computable action $G \curvearrowright X \subseteq \{0, 1\}^{\mathbb{N}}$ is a topological factor of a G -subshift of finite type.*

Before sketching the proof, let us record the object which the grids will carry. Fix finite generating sets S_H and S_V and let

$$S = (S_H \times \{1_{G_V}\}) \cup (\{1_{G_H}\} \times S_V),$$

a generating set of G which we assume contains 1_G .

Definition 2.38 (Set representation). Let $G \curvearrowright X \subseteq \{0, 1\}^{\mathbb{N}}$ and $\Sigma' = \{0, 1\}^S$. For $y \in \Sigma'^{\mathbb{N}}$ and $s \in S$ write $\pi_s y = (y_n(s))_{n \in \mathbb{N}} \in \{0, 1\}^{\mathbb{N}}$. The *set representation* of the action is

$$Y = \{y \in \Sigma'^{\mathbb{N}} : \pi_{1_G} y \in X \text{ and } \pi_s y = s \cdot (\pi_{1_G} y) \text{ for every } s \in S\}.$$

A point of Y is a point of X together with its images under all the generators, all written on a single sequence. The action is computable precisely when Y is an effectively closed subset of $\Sigma'^{\mathbb{N}}$, which is the form in which the computation tiling space of [Section 2.4](#) can use it.

Proof sketch of Theorem 2.37. Let $G \curvearrowright X$ be a computable action and let Y be its set representation, an effectively closed subset of $\Sigma'^{\mathbb{N}}$. Let $\tilde{P}$ and the maps $\gamma_g: \mathbb{N}^2 \rightarrow G$ be as in [Corollary 2.31](#), and let $T_{\text{comp}, Y}$ be the computation tiling space of Y , with tile alphabet B .

Define $Z \subseteq \tilde{P} \times B^G \times \Sigma'^G$ as the set of configurations $z = (\rho, \tau, \sigma)$ satisfying the following two families of local rules.

Embedding rule. For every $g \in G$, the tiles and symbols seen along the grid of g obey the rules of $T_{\text{comp}, Y}$. Concretely, the position $\gamma_g((0, 0), \rho)$ is required to carry the tile *seed*; the right neighbor of g inside its grid is $gL_{\bar{t}}(\rho_H(g))$ and its upper neighbor is $gL_{\bar{t}}(\rho_V(g))$, and one imposes that the right side of $\tau(g)$ matches the left side of the tile at the former, that the upper side of $\tau(g)$ matches the bottom side of the tile at the latter, that σ takes the same value at both, which is the synchronization condition, and finally the compatibility condition between τ and σ from the definition of $T_{\text{comp}, Y}$.

Coherence rule. For every $g \in G$ and every generator $s \in S$, the s -component of the symbol of σ read at the grid of g agrees with the 1_G -component of the symbol read at the grid of gs^{-1} , at the corresponding position.

All of these conditions involve only finitely many coordinates of z around g , since all displacements lie in $K_H \cup K_V$ and S is finite. As $\tilde{P}$ is an SFT, Z is a G -subshift of finite type.

Unwinding the definitions, a configuration $z = (\rho, \tau, \sigma) \in Z$ satisfies:

1. for every $g \in G$, $(\tau(\gamma_g(u, \rho)), \sigma(\gamma_g(u, \rho)))_{u \in \mathbb{N}^2} \in T_{\text{comp}, Y}$, that is, the restriction of z to the grid of g is a valid seeded computation tiling;
2. for every $g \in G$, $s \in S$ and $u \in \mathbb{N}^2$, $\sigma(\gamma_g(u, \rho))(s) = \sigma(\gamma_{gs^{-1}}(u, \rho))(1_G)$.

The first item holds because the two neighbors used in the embedding rule are exactly the two neighbors of g inside its grid, so the rule transports the local rules of the tiling space along γ_g . The second follows from the coherence rule together with the fact that, by synchronization, it is enough to check the identity along one of the two axes.

Now define $\phi: Z \rightarrow \{0, 1\}^{\mathbb{N}}$ by

$$\phi(\rho, \tau, \sigma)_n = \sigma(\gamma_{1_G}((n, 0), \rho))(1_G).$$

This map is continuous, since each coordinate depends on finitely many coordinates of z . It takes values in X : by item (1) and [Proposition 2.36](#) the sequence read along the bottom row of the grid of 1_G belongs to Y , hence its 1_G -component belongs to X . It is equivariant: using item (2) and (1),

$$s \cdot \phi(z) = (\sigma(\gamma_{1_G}((n, 0), \rho))(s))_n = (\sigma(\gamma_{s^{-1}}((n, 0), \rho))(1_G))_n = \phi(s \cdot z),$$

and S generates G .

Finally, ϕ is surjective. Given $x \in X$, choose $y \in Y$ with $\pi_{1_G} y = x$ and fix $\rho \in \tilde{P}$. For each $g \in G$, [Proposition 2.36](#) provides a computation tiling $(\tau^g, \sigma^g) \in T_{\text{comp}, Y}$ whose bottom row is $g^{-1} \cdot y$. Since the grids are pairwise disjoint by [Corollary 2.31](#), we may write (τ^g, σ^g) on the grid of g , for every g simultaneously, and fill the positions covered by no grid with an arbitrary fixed symbol. The resulting configuration lies in Z : the embedding rule holds grid by grid, and the coherence rule holds because on the grid of g one reads $g^{-1} \cdot y$ while on the grid of gs^{-1} one reads $sg^{-1} \cdot y$, and $(g^{-1}y)_m(s) = (sg^{-1}y)_m(1_G)$ by definition of the set representation. By construction ϕ maps it to x . $\square$

Remark 2.39. The construction uses no hierarchical or self-similar structure, in contrast with the simulation theorems of [Section 2.1](#). Everything rests on the two ingredients we have isolated: the paradoxical subshift provides, by local rules, a family of disjoint $\mathbb{N}^2$ -grids indexed by the group, and Wang tiles provide, again by local rules, the space-time diagram of a Turing machine on each grid. The direct product is what turns the one-dimensional paths of one factor into two-dimensional grids.

Example 2.40. The groups $\mathbb{F}_k \times \mathbb{F}_\ell$ for $k, \ell \geq 2$ are self-simulable. These were the first known examples. Note that each factor $\mathbb{F}_k$ is itself not self-simulable, as it has infinitely many ends; we shall return to this obstruction in [Section 3](#).

2.6 Stability properties and examples

[Theorem 2.37](#) produces self-simulable groups, but every one of them is a direct product. The rest of [\[BSS26\]](#) is devoted to propagating self-simulability to groups which are not presented as products. We state the three tools without proof and then review the examples they yield.

The first tool transfers self-simulability from a subgroup, provided that adjacent cosets have enough room to communicate. Recall that the proof of [Theorem 2.37](#) wrote a copy of the simulated point on a family of disjoint paths; if $\Delta \leq G$ and we want to run this argument on each coset of Δ , we need that a path in Δ can be pushed by a generator t to a path in $t\Delta$. This is what the following algebraic condition guarantees.

Definition 2.41. A subgroup $\Delta \leq G$ is *mediated* if there exists a finite generating set T of G such that $\Delta \cap t\Delta t^{-1}$ is non-amenable for every $t \in T$.

Theorem 2.42 ([\[BSS26, Theorem 1.3\]](#)). *Let G be a finitely generated and recursively presented group which contains a mediated self-simulable subgroup. Then G is self-simulable.*

Corollary 2.43 ([\[BSS26, Corollary 6.12\]](#)). *If G is finitely generated, recursively presented, and admits a normal self-simulable subgroup, then G is self-simulable. In particular, if N is self-simulable then so is $N \times H$ for every finitely generated and recursively presented H .*

Merely containing a self-simulable subgroup is not enough: the group $(\mathbb{F}_2 \times \mathbb{F}_2) * \mathbb{F}_2$ contains $\mathbb{F}_2 \times \mathbb{F}_2$ but has infinitely many ends, and we shall see in [Section 3](#) that this forbids self-simulability. Neither does admitting a self-simulable *quotient* help: $\mathbb{F}_4$ surjects onto $\mathbb{F}_2 \times \mathbb{F}_2$ and yet is not self-simulable, again because of its ends.

The other two tools are rigidity statements. We recall the notion of quasi-isometry:

Definition 2.44. Two graphs $G = (V, E)$ and $G' = (V', E')$ are *quasi-isometric* if there exists a function $f: V \rightarrow V'$ and positive real constants A , B and C such that

- f is a *quasi-isometric embedding*: for every $v_1, v_2 \in V$ one has

$$\frac{1}{A}d_G(v_1, v_2) - B \leq d_{G'}(f(v_1), f(v_2)) \leq Ad_G(v_1, v_2) + B.$$

- $f(V)$ is *relatively dense*: for every $v' \in V'$ there exists $v \in V$ such that

$$d_{G'}(v', f(v)) \leq C.$$

A quasi-isometry is an equivalence relation that captures the large scale geometry of an object. Note that all Cayley graphs (given by a generating set) of a fixed group are quasi-isometric, thus we may speak about quasi-isometries between groups.

Theorem 2.45 ([BSS26, Theorems 1.4 and 1.5]). *Self-simulability is an invariant of commensurability among finitely generated groups, and an invariant of quasi-isometry among finitely presented groups.*

The quasi-isometry statement is the deepest of the three. Since amenability is itself a quasi-isometry invariant, and amenable groups are never self-simulable, one may assume both groups non-amenable, and then a theorem of Whyte [Why99] turns the quasi-isometry into a bilipschitz bijection. The bilipschitz map is then encoded inside an SFT on either group by a construction of Cohen [Coh17].

Groups quasi-isometric to a product of trees

The simplest application of Theorem 2.45 concerns the finitely presented simple groups constructed by Burger and Mozes [BM00]. These groups act geometrically, that is, properly and cocompactly by isometries, on a product of two regular trees of degree at least three. By the Švarc–Milnor lemma any group acting geometrically on a metric space is quasi-isometric to that space, so a Burger–Mozes group is quasi-isometric to $T_1 \times T_2$, hence to $\mathbb{F}_k \times \mathbb{F}_k$ for suitable $k \geq 2$, which is self-simulable by Theorem 2.37. As Burger–Mozes groups are finitely presented, Theorem 2.45 applies.

Corollary 2.46 ([BSS26, Corollary 8.2]). *The groups of Burger and Mozes are self-simulable. In particular there exist finitely presented non-amenable simple groups which are self-simulable.*

Thompson’s groups

The next family is obtained from a criterion which asks only for a sufficiently rich action on a zero-dimensional space.

Lemma 2.47 ([BSS26, Lemma 8.3]). *Let G be a finitely generated and recursively presented group acting faithfully by homeomorphisms on a zero-dimensional space X . Suppose that for every nonempty open set $U \subseteq X$ the subgroup G_U of elements which act trivially on $X \setminus U$ is non-amenable. Then G is self-simulable.*

Idea of the proof. Since X is zero-dimensional, write $X = A \sqcup B$ with A, B nonempty and clopen. Faithfulness gives $G_A \cap G_B = \{1_G\}$, and elements supported on disjoint sets commute, so G_A and G_B generate a direct product inside G . Both are non-amenable by hypothesis, so they contain finitely generated non-amenable subgroups Δ_A and Δ_B , and $\Delta_A \times \Delta_B$ is self-simulable by Theorem 2.37. It remains to enlarge this product into a mediated subgroup: for each generator t of G one picks a clopen $U_t \subseteq A$ small enough that tU_t is contained in A or in B , and uses a finitely generated non-amenable subgroup of G_{tU_t} as the witness of non-amenableity required by the definition of mediation. Theorem 2.42 then applies. $\square$

Recall Thompson’s groups $F \leq T \leq V$. The group V is most conveniently described as a group of homeomorphisms of the Cantor space $\mathcal{C} = \{0, 1\}^{\mathbb{N}}$: an element of V is given by two partitions $\mathcal{C} = [w_1] \sqcup \cdots \sqcup [w_n] = [u_1] \sqcup \cdots \sqcup [u_n]$ into cylinders, and acts by replacing the prefix w_i by u_i . The groups F and T are the subgroups obtained by requiring the sequences (w_i) and

(u_i) to be in lexicographic order, respectively in lexicographic order up to a cyclic permutation; equivalently, F and T are the groups of piecewise linear homeomorphisms of $[0, 1]$, respectively of $\mathbb{R}/\mathbb{Z}$, whose slopes are powers of 2 and whose breakpoints are dyadic. See [CFP96]. The group V was, to the best of our knowledge, the first infinite finitely presented simple group. Brin’s higher-dimensional groups nV act in the same way on $\mathcal{C}^n$, see [Bri04].

Non-amenability of V , and of T , is easy to see: the natural action on $\mathcal{C}$ admits no invariant Borel probability measure, since all cylinders would be forced to have the same measure, so amenability would contradict the Krylov–Bogolyubov theorem. The key structural point for us is that the subgroups fixing the complement of a cylinder are large: for V and nV the subgroup of elements acting trivially outside a cylinder is isomorphic to the whole group, and for $\Gamma \in \{F, T\}$ and any nontrivial word w the subgroup of Γ acting trivially outside $[w]$ is isomorphic to F . Lemma 2.47 therefore gives the following.

Corollary 2.48 ([BSS26, Corollaries 8.4 and 8.5]). *Thompson’s group V , and more generally the Brin–Thompson groups nV , are self-simulable. Thompson’s group F is self-simulable if and only if F is non-amenable, and if F is non-amenable then T is self-simulable as well.*

The statement about F deserves a comment, since the amenability of F is a famous open problem. One direction is Lemma 2.47, applied to the action of F on $\mathcal{C}$ under the assumption that F is non-amenable. The other is the obstruction of Section 3: a finitely generated, recursively presented amenable group is never self-simulable, because it admits computable actions of infinite topological entropy while the entropy of a subshift is finite, and entropy does not increase under factor maps. Combining the two directions turns a question about amenability into a question about computability.

Theorem 2.49 ([BSS26, Theorem 1.7]). *Thompson’s group F is amenable if and only if there exists a computable action of F which is not the topological factor of an F -subshift of finite type.*

One further consequence of Corollary 2.48 is rather funny: the natural action of V on $\{0, 1\}^{\mathbb{N}}$ is expansive and computable, hence topologically conjugate to an effective subshift, and therefore to a sofic one. Hence we may say it is “sofic” (it is still not known whether Thompson’s V is a sofic group though).

Further examples

The remaining examples in [BSS26, Section 8] follow from the same tools, usually through an abstract orthogonality criterion generalizing Lemma 2.47: one exhibits a family of subgroups indexed by a semilattice with an orthogonality relation, such that the subgroups attached to orthogonal indices commute and intersect trivially, enough of them are non-amenable, and the generators of the group are supported on indices which avoid at least one of two fixed orthogonal ones. The list obtained is:

- finitely generated non-amenable branch groups [BGv03], which are commensurable to a direct product of at least two copies of a group, hence covered by Theorem 2.37 together with the commensurability invariance of Theorem 2.45;
- the general and special linear groups $\mathrm{GL}_n(\mathbb{Z})$ and $\mathrm{SL}_n(\mathbb{Z})$ for $n \geq 5$, using that they are generated by transvections, which are supported on pairs of coordinates, and that $\mathrm{SL}_2(\mathbb{Z})$ is non-amenable;
- the automorphism and outer automorphism groups $\mathrm{Aut}(\mathbb{F}_n)$ and $\mathrm{Out}(\mathbb{F}_n)$ for $n \geq 5$, using that they are generated by Nielsen transformations;

- the braid groups B_n for $n \geq 7$, using the nesting $B_m \leq B_n$ on contiguous strands;
- right-angled Artin groups $A(\mathcal{G})$ where the complement of the defining graph $\mathcal{G}$ is connected and contains two edges no vertex is adjacent to both of, which holds for instance when the complement has diameter at least five, or is a cycle of length at least seven, see [Cha07].

We remark that self-simulability on Artin groups was recently completely characterized by Blot and Salo [BS26].

It was asked in [BSS26] whether every one-ended word-hyperbolic group is self-simulable; we shall see in Section 3 that the answer is negative, already for the fundamental group of a closed orientable surface of genus two. It remains open whether $\mathrm{GL}_3(\mathbb{Z})$ and $\mathrm{GL}_4(\mathbb{Z})$ are self-simulable, and whether $G \times \mathbb{Z}$ being self-simulable forces G to be.

3 Lecture III: Obstructions to self-simulation

Which groups are *not* self-simulable? By definition, an obstruction is an effective subshift, or more generally a computable action, which is not the topological factor of any SFT. This lecture is devoted to a single such obstruction, of a geometric nature, which covers every example known in the literature. It comes from [BBSS27].

3.1 The shape of an obstruction

There is one easy obstruction, which we have already used in Section 2.6 and which we record for completeness. If G is amenable, every subshift on G has finite topological entropy, while a recursively presented group always admits computable actions of infinite topological entropy: the action of G on the set of configurations $x \in \{0, 1\}^{S^* \times \mathbb{N}}$ satisfying $x(w, n) = x(w', n)$ whenever $\overline{w} = \overline{w'}$, which is a full shift on an infinite alphabet in disguise, is one. As entropy does not increase under factor maps, no recursively presented amenable group is self-simulable; see [BSS26, Proposition 3.7].

This argument uses non-expansive actions in an essential way, and it says nothing about subshifts. The question we address now is the sharper one: for which groups does there exist an *effective subshift* which is not sofic? The model example is the *mirror shift* on $\mathbb{Z}^2$, described by Hochman [Hoc16]: configurations over $\{0, 1, \star\}$ in which, if a $\star$ occurs, then it occupies an entire vertical line and the configuration is symmetric with respect to that line, see Figure 5. It is effective, and it is not sofic because an SFT cover would have to transmit, across the line of $\star$'s, more information than the finitely many symbols available on a column of bounded width allow. The same idea was used in [ABS17] for amenable and for multi-ended groups, and in [BSS25] for $\mathbb{F}_2 \times \mathbb{Z}$.

What the argument really needs is not the symmetry, and not the group, but a geometric configuration: two sets of the same size, close to each other, which become far apart once a comparatively small set is deleted. This is what the next definition captures.

3.2 UFOs and extraterrestrial groups

Let $\mathcal{G} = (V, E)$ be a graph. Given finite $A, B \subseteq V$, a *matching* between A and B is a relation $M \subseteq A \times B$ whose two projections are injective; it is *complete* if it induces a bijection between A and B , and it is *by paths of length at most k* if $d_{\mathcal{G}}(u, v) \leq k$ for every $(u, v) \in M$.

Definition 3.1 (UFO). Let $\mathcal{G} = (V, E)$ be a graph and let $m, k, r \in \mathbb{N}$. An (m, k, r) -UFO is a triple (U, F, O) of pairwise disjoint finite subsets of V such that

1. $|U| \geq m|F|$;

0	0	0	0	0	0	0	0	★	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	★	0	0	0	1	0	0	0
0	0	0	1	1	0	0	0	★	0	0	1	1	0	0	0
0	0	0	0	1	1	1	1	★	1	1	1	0	0	0	0
0	0	0	1	1	1	1	1	★	1	1	1	1	0	0	0
0	0	1	1	0	0	1	1	★	1	0	0	1	1	0	0
0	0	1	1	0	0	1	1	★	1	0	0	1	1	0	0
0	1	1	1	1	1	1	1	★	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	★	1	1	1	1	1	1	0
0	0	0	1	0	1	0	0	★	0	1	0	1	0	0	0
0	0	0	1	0	1	0	0	★	0	1	0	1	0	0	0
0	0	1	0	1	0	0	0	★	0	0	1	0	1	0	0
0	0	0	0	0	0	0	0	★	0	0	0	0	0	0	0

Figure 5: A portion of a configuration in the mirror shift.

2. there is a complete matching between U and O by paths of length at most k ;
3. every path from U to O either passes through a vertex of F or has length at least r .

Definition 3.2 (Extraterrestrial graph). A graph $\mathcal{G}$ is *extraterrestrial* if for every $m \in \mathbb{N}$ there is $k \in \mathbb{N}$ such that for every $r \in \mathbb{N}$ the graph $\mathcal{G}$ admits an (m, k, r) -UFO.

The letters stand for *Under*, *Front* and *Over*, see Figure 6. In the canonical Cayley graph of $\mathbb{Z}^2$ one pictures U as a block of vertices below a long horizontal wall F , and O as the mirror block above it. The quantifiers matter and should be read carefully. The parameter m says how large U is compared to the wall F ; k , which is chosen after m but before r , bounds the distance between matched pairs; and r , which may be taken arbitrarily large afterwards, says how far apart U and O become once the wall is removed.

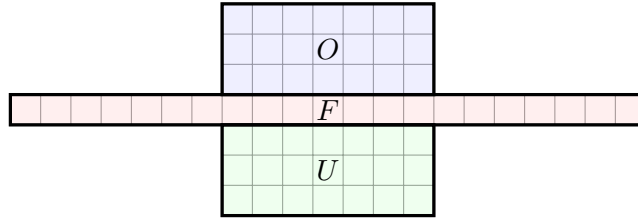


Figure 6: A $(1, 4, 18)$ -UFO for the canonical Cayley graph of $\mathbb{Z}^2$.

Example 3.3 ($\mathbb{Z}^d$). Let S be the canonical generating set of $\mathbb{Z}^d$. For $m, r \geq 1$ put

$$U = [0, r-1]^{d-1} \times [-3^{d-1}m, -1], \quad F = [-r, 2r-1]^{d-1} \times \{0\}, \quad O = [0, r-1]^{d-1} \times [1, 3^{d-1}m].$$

Then (U, F, O) is an $(m, 3^{d-1}m + 1, 2r + 4)$ -UFO: the sets U and O are matched by vertical translation, while a path from U to O avoiding F must leave the slab $[-r, 2r-1]^{d-1} \times \mathbb{Z}$ before crossing the hyperplane, which costs at least $2r$ steps. Hence $\mathbb{Z}^d$ is extraterrestrial.

Theorem 3.4 ([BBSS27, Theorem A]). *Being extraterrestrial is a quasi-isometry invariant among graphs of bounded degree.*

The proof is straightforward but tedious, so we leave it as a (very annoying) exercise: one pushes a UFO (U, F, O) of $\mathcal{G}$ forward along a quasi-isometry f , thickening F into the set F' of vertices at distance at most α from $f(F)$ for a constant α depending on the quasi-isometry constants and on the maximal degree, and discarding from the matching the pairs which collide or fall into F' . Each of the three conditions survives with modified parameters.

Since two Cayley graphs of the same finitely generated group with respect to two finite generating sets are quasi-isometric, we may speak of the group rather than of the graph.

Definition 3.5. A finitely generated group G is *extraterrestrial* if $\text{Cay}(G, S)$ is extraterrestrial for some, equivalently every, finite symmetric generating set S . Likewise, for $H \leq G$ we say that the pair (H, G) is extraterrestrial if the Schreier graph $\text{Sch}(G, H, S)$ is.

3.3 The generalized mirror shift

Theorem 3.6 ([BBSS27, Theorem B]). *Let G be a finitely generated extraterrestrial group. Then there exists a G -subshift which is effectively closed by patterns and which is not the topological factor of any G -SFT. In particular, if G is recursively presented then G is not self-simulable.*

We describe the construction and the proof in the case where G has decidable word problem, which is where the geometry is visible, and indicate at the end what has to be added in general.

Fix a finite symmetric generating set S and a well-order $1_G < g_1 < g_2 < \dots$ of G . Let $A = \{\star, \mathbf{u}, \mathbf{o}\}$. For $x \in A^G$ write $U(x) = \{g : x(g) = \mathbf{u}\}$ and $O(x) = \{g : x(g) = \mathbf{o}\}$, and let $M(x) \subseteq U(x) \times O(x)$ be the *greedy matching*: first match every still unmatched pair of the form (h, hg_1) , then every still unmatched pair of the form (h, hg_2) , and so on. The point of the greedy rule is that it is determined locally, in the sense that whether (h, hg_i) is matched depends only on the symbols within distance $|g_i|_S$ of h , and hence can be enforced by a recursively enumerable set of forbidden patterns.

Definition 3.7 (Generalized mirror shift). Let $\Sigma = A \times \{-, +\}$ and let $X_{\text{GM}} \subseteq \Sigma^G$ be the set of pairs $(x, y) \in A^G \times \{-, +\}^G$ such that

$$y(g) = y(g') \quad \text{for every } (g, g') \in M(x),$$

that is, matched positions must carry the same sign.

If the word problem of G is decidable then the greedy matching is computable and X_{GM} is an effective subshift. Note how this generalizes the mirror shift: there, the matching was given by the reflection across the line of $\star$'s; here, it is given by an abstract greedy rule, and the geometry which made the reflection work is replaced by the existence of UFOs.

Proof sketch of Theorem 3.6. Suppose X_{GM} were sofic. Then there are an S -nearest neighbor SFT Z with alphabet of size q and a 1-block factor map $\phi: Z \rightarrow X_{\text{GM}}$. Fix an integer $m > 2 \log_2 q$ and, using that G is extraterrestrial, a k such that for every r there is an (m, k, r) -UFO (U_r, F_r, O_r) .

Step 1: a counting argument. For each r consider the finite set W_r of configurations of X_{GM} whose A -layer equals $\mathbf{u}$ on U_r , $\mathbf{o}$ on O_r and $\star$ elsewhere. All of them induce the same matching M_r , so the elements of W_r differ only in the signs carried by $U_r \cup O_r$, and these signs are free except that they must agree along M_r . Let $M_{r,k}$ be the set of pairs of M_r matched by paths of length at most k . Because the UFO provides a complete matching of U_r with O_r by paths of length at most k , and because the greedy rule matches by increasing order, no two elements of such a pair can both be left unmatched by $M_{r,k}$; hence $2|M_{r,k}| \geq |U_r|$. Therefore the number of admissible sign patterns is at least

$$2^{|M_{r,k}|} \geq 2^{|U_r|/2} \geq 2^{m|F_r|/2} > q^{|F_r|},$$

the last inequality by the choice of m . The right-hand side is the number of patterns that a configuration of Z can display on F_r . By the pigeonhole principle there are $z^1, z^2 \in Z$ which agree on F_r and whose images differ in the sign of some pair $(g, g') \in M_{r,k}$.

Step 2: a limit. Translate z^1 and z^2 by g^{-1} , so that the disagreeing pair is $(1_G, h)$ with $|h|_S \leq k$, and let (z, z') be a limit point of the resulting sequence as $r \rightarrow \infty$. By compactness, $\phi(z)$ and $\phi(z')$ have the same A -layer, hence the same matching; they disagree on the sign of one matched pair $(1_G, h)$; and the set F of positions where z and z' agree separates $U(\phi(z))$ from $O(\phi(z))$ in $\text{Cay}(G, S)$. This last point is exactly where the third condition of [Definition 3.1](#) is used: a path joining the two sides and avoiding F would, for r large, give a path from U_r to O_r of bounded length avoiding F_r .

Step 3: splicing. Let W be the union of the connected components of $U(\phi(z))$ in $\text{Cay}(G, S) \setminus F$, and define z^* to be z on W and z' elsewhere. Every edge leaving W lands in F , where z and z' agree, so no nearest neighbor rule of Z is violated and $z^* \in Z$. As ϕ is a 1-block map, $\phi(z^*)$ has the same A -layer, hence the same matching, but it carries the sign of $\phi(z)$ at 1_G and the sign of $\phi(z')$ at h . These differ, so $\phi(z^*) \notin X_{\text{GM}}$, a contradiction. $\square$

Remark 3.8. For a general finitely generated group the greedy matching need not be computable, since deciding whether $h = gg_i$ is the word problem. The fix in [\[BBSS27\]](#) is to work on the free group $F(S)$ and add a layer $\{0, 1\}^{F(S)}$ which *guesses* the word problem: two elements are declared equal when their $\{0, 1\}$ -neighborhoods of a suitable radius agree, and one forbids patterns where this pseudo-equality is inconsistent with the rest of the configuration. A result of Aubrun, Barbieri and Thomassé, obtained from a probabilistic argument on colorings of bounded degree graphs, guarantees that some configuration realizes the true word problem in this way, with a radius linear in the distance. The resulting subshift is effectively closed by patterns, which is the right notion for groups which are not recursively presented.

3.4 Which groups are extraterrestrial?

We now give the three basic examples in full, and then survey the rest.

Proposition 3.9 ([\[BBSS27, Proposition 5.1\]](#)). *Every infinite finitely generated amenable group is extraterrestrial.*

Proof. Let S be a finite symmetric generating set containing 1_G and let $m \geq 1$. By [Lemma 2.12](#) applied with $K = S$ and $\varepsilon = 1/m$ there is a nonempty $U \subseteq G$ with $|US| \leq (1 + \frac{1}{m})|U|$, that is,

$$|US \setminus U| \leq \frac{1}{m}|U|.$$

Put $F = US \setminus U$. Since G is infinite and US is finite, the set $G \setminus US$ is infinite, so we may choose $O \subseteq G \setminus US$ with $|O| = |U|$. Then $|U| \geq m|F|$, which is condition (1).

As $|U| = |O|$, any bijection between them is a complete matching, and taking $k = \max\{d_S(g, g') : g \in U, g' \in O\}$, which is finite, it is a matching by paths of length at most k . This is condition (2).

For condition (3), let $g_0, \dots, g_\ell$ be a path in $\text{Cay}(G, S)$ with $g_0 \in U$ and $g_\ell \in O$. Since $g_\ell \notin US \supseteq U$, there is a first index i with $g_i \notin U$. Then $g_i = g_{i-1}s$ for some $s \in S$ with $g_{i-1} \in U$, so $g_i \in US \setminus U = F$. Hence *every* path from U to O meets F , and condition (3) holds vacuously for every r . Thus (U, F, O) is an (m, k, r) -UFO for every $r \in \mathbb{N}$; see [Figure 7](#). $\square$

Proposition 3.10 ([\[BBSS27, Proposition 5.2\]](#)). *Every finitely generated multi-ended group is extraterrestrial.*

Proof. By definition there are a finite generating set S and a finite set $F \subseteq G$ such that $\text{Cay}(G, S) \setminus F$ has at least two infinite connected components, say C_1 and C_2 . Let $m \geq 1$. As C_1 is infinite we may choose $U \subseteq C_1$ with $|U| \geq m|F|$, and as C_2 is infinite we may choose $O \subseteq C_2$ with $|O| = |U|$; this gives condition (1). Any bijection between U and O is a complete matching, and it is by paths of length at most $k = \max\{d_S(g, g') : g \in U, g' \in O\}$, which gives

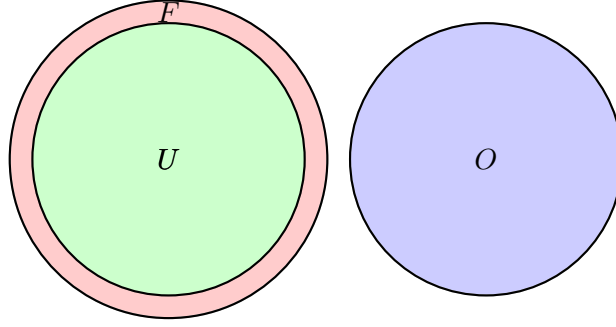


Figure 7: Amenable groups are extraterrestrial.

condition (2). Finally, U and O lie in distinct connected components of $\text{Cay}(G, S) \setminus F$, so no path from U to O avoids F and condition (3) holds for every r . Note that the same F , and hence the same k , works for all r ; see Figure 8. $\square$

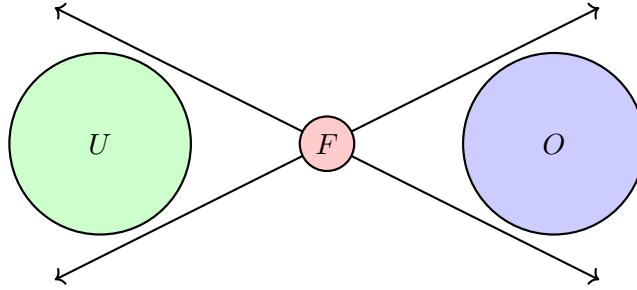


Figure 8: Multi-ended groups are extraterrestrial.

Both proofs produce a UFO which works for every r at once, because the wall F genuinely disconnects U from O . The next example is the interesting one: there the wall does not disconnect anything, and the parameter r has to be used.

Definition 3.11 (Pentagon model). The *pentagon model* is the graph with vertex set $\mathbb{Z}^2$ and edges

$$\{(h, v), (h+1, v)\} : (h, v) \in \mathbb{Z}^2\} \cup \{(h, v), (2h, v-1)\} : (h, v) \in \mathbb{Z}^2\}.$$

The map $(h, v) \mapsto h2^v + 2^v i$ is a quasi-isometry from the pentagon model onto the hyperbolic plane $\mathbb{H} = \{z \in \mathbb{C} : \text{Im}(z) > 0\}$ with its usual metric: the horizontal edges are the horocyclic steps and the diagonal edges move between consecutive horocycles.

Lemma 3.12 ([BSS27, Lemma 5.17]). *The pentagon model is extraterrestrial.*

Proof. Let $m, r \geq 1$ and put

$$\begin{aligned} U &= \{(h, v) : -3m \leq h < 0, 0 \leq v < r\}, \\ F &= \{(0, v) : -r \leq v \leq 2r-1\}, \\ O &= \{(h, v) : 0 < h \leq 3m, 0 \leq v < r\}. \end{aligned}$$

Then $|U| = |O| = 3mr$ and $|F| = 3r$, so $|U| \geq m|F|$ and condition (1) holds. The relation $M = \{((-h, v), (h, v)) : 0 < h \leq 3m, 0 \leq v < r\}$ is a complete matching between U and O , and $(-h, v)$ and (h, v) are joined by the horizontal path which increases the first coordinate one step at a time, of length $2h \leq 6m$; so condition (2) holds with $k = 6m + 1$, a value which does not depend on r .

For condition (3), observe that both types of edge preserve the sign of the first coordinate: a horizontal edge changes it by one unit, and a diagonal edge maps h to $2h$. Hence a path from U , where $h < 0$, to O , where $h > 0$, must visit a vertex whose first coordinate is 0. If the path avoids F , that vertex has second coordinate either $< -r$ or $> 2r - 1$. As every edge changes the second coordinate by at most one, and the path starts at a vertex with $0 \leq v < r$, the path has length at least r . Thus (U, F, O) is an $(m, 6m + 1, r)$ -UFO, see Figure 9, and as m and r were arbitrary the pentagon model is extraterrestrial. $\square$

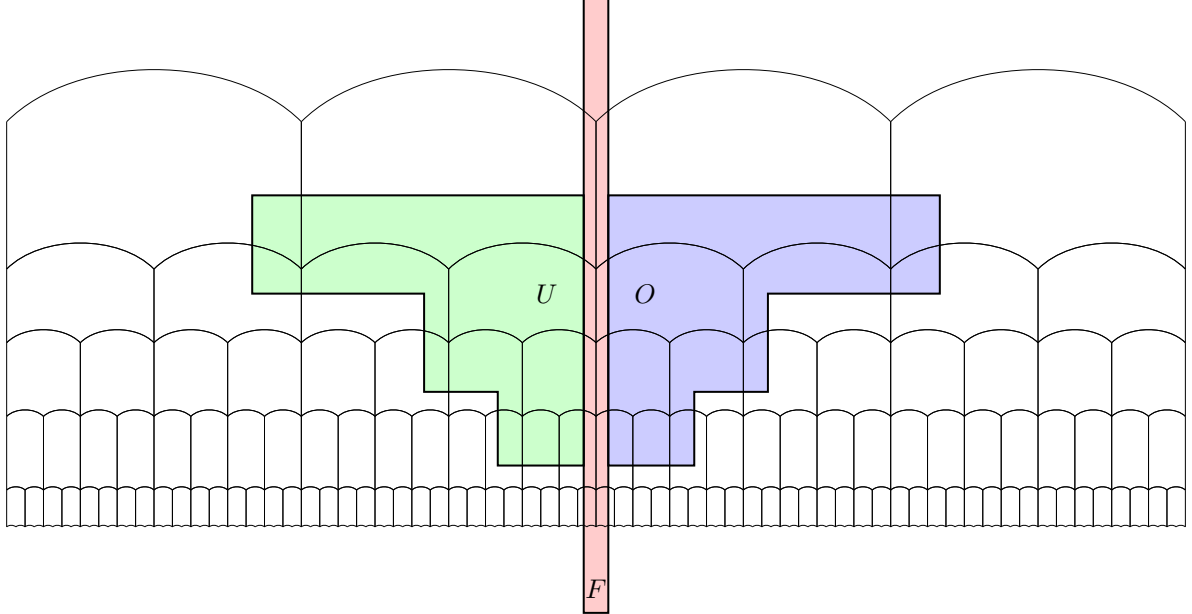


Figure 9: The pentagon model is extraterrestrial.

Proposition 3.13. *Every finitely generated group quasi-isometric to the hyperbolic plane is extraterrestrial. In particular every cocompact Fuchsian group is extraterrestrial.*

Proof. The pentagon model is quasi-isometric to $\mathbb{H}$, hence to the Cayley graph of any such group, and Theorem 3.4 together with Lemma 3.12 gives the first claim. A cocompact Fuchsian group is a discrete subgroup of $\text{PSL}(2, \mathbb{R})$ acting cocompactly on $\mathbb{H}$, so it is quasi-isometric to $\mathbb{H}$ by the Švarc–Milnor lemma. $\square$

Corollary 3.14 ([BSS27, Corollary 5.20]). *The fundamental group of any surface with finite topology is extraterrestrial. In particular, no such group is self-simulable.*

Proof. If the surface is not closed, its fundamental group is free, hence multi-ended when it is nontrivial, and Proposition 3.10 applies. A non-orientable closed surface has an orientable double cover, so its fundamental group is commensurable with, hence quasi-isometric to, that of an orientable one, and we may assume the surface closed and orientable of genus g . If $g \leq 1$ the fundamental group is trivial or $\mathbb{Z}^2$, hence amenable, and Proposition 3.9 applies. If $g \geq 2$ then

$$\pi_1(\Sigma_g) = \langle a_1, b_1, \dots, a_g, b_g \mid [a_1, b_1] \cdots [a_g, b_g] = 1 \rangle$$

acts cocompactly on $\mathbb{H}$, so Proposition 3.13 applies. $\square$

This is worth emphasizing, since it settles a question left open in [BSS26]: the fundamental group of a closed orientable surface of genus 2 is one-ended, hyperbolic, non-amenable and finitely presented, and it is *not* self-simulable. So neither one-endedness nor non-amenableity, nor both together, suffice.

3.5 A survey of extraterrestrial groups

The proofs of [Proposition 3.9](#) and [Proposition 3.10](#) have a common generalization, which is the main source of examples.

Theorem 3.15 ([BBSS27, Theorem 5.4 and Corollary 5.6]). *Let G be finitely generated and let $H \leq G$ be an amenable subgroup. If the Schreier graph $\text{Sch}(G, H, S)$ is extraterrestrial, then so is G . In particular this holds when $\text{Sch}(G, H, S)$ is multi-ended, in which case we say that G admits an amenable cut.*

The proof takes a UFO in the Schreier graph, lifts it to G by choosing one representative per coset, and then thickens the lift by a Følner set of H , which is what keeps the wall small relative to U . Every class listed below, except for the ones obtained geometrically, is an instance of it:

- short exact sequences $1 \rightarrow H \rightarrow G \rightarrow K \rightarrow 1$ with H amenable and K extraterrestrial; in particular $H \times K$ with H amenable and K not one-ended, which recovers the fact that $\mathbb{F}_2 \times \mathbb{Z}$ is not self-simulable;
- amalgamated free products $G_1 *_H G_2$ over an amenable H with both embeddings proper, and HNN extensions $G *_\phi$ over an amenable subgroup;
- consequently, by Bass–Serre theory, any finitely generated group acting cocompactly and without edge inversions on a tree, with no globally fixed vertex and with amenable edge stabilizers; in particular all generalized Baumslag–Solitar groups;
- every hyperbolic group with infinite outer automorphism group, since by a theorem of Paulin [Pau91] such a group splits over a virtually cyclic subgroup;
- cocompact Fuchsian groups and fundamental groups of surfaces of finite topology, as above; for the genus two surface group one can also exhibit an amenable cut directly, taking $H = \pi_1(X) \cong \mathbb{Z}$ for an incompressible subsurface X with two boundary components.

Finally, the class interacts with the open problem we met in [Section 2.6](#).

Proposition 3.16 ([BBSS27, Proposition 5.3]). *Thompson’s group F is amenable if and only if it is extraterrestrial.*

Proof. If F is amenable it is extraterrestrial by [Proposition 3.9](#). Conversely, if F is extraterrestrial then, being finitely presented and hence recursively presented, [Theorem 3.6](#) shows that it is not self-simulable, and [Corollary 2.48](#) then gives that F is amenable. $\square$

In the two questions below, *strongly self-simulable* is the variant of self-simulability adapted to groups which are not recursively presented, obtained by relativizing effectiveness to the free group on a generating set; see [BBSS27, Section 2.2]. Two questions close the picture, and the course. The first is whether the obstruction is complete.

Question 3.17 ([BBSS27, Question 1.1]). Is there a finitely generated group which is neither strongly self-simulable nor extraterrestrial?

Question 3.18 ([BBSS27, Question 1.2]). Is there an extraterrestrial group which does not admit an amenable subgroup whose Schreier graph is multi-ended?

No example is known in either direction: every group which we can prove not to be self-simulable is extraterrestrial, and every extraterrestrial group which we know of has an amenable cut.

References

- [ABJ18] Nathalie Aubrun, Sebastián Barbieri, and Emmanuel Jeandel, *About the domino problem for subshifts on groups*, Sequences, Groups, and Number Theory, Trends in Mathematics, Birkhäuser, 2018, pp. 331–389.
- [ABS17] Nathalie Aubrun, Sebastián Barbieri, and Mathieu Sablik, *A notion of effectiveness for subshifts on finitely generated groups*, Theoret. Comput. Sci. **661** (2017), 35–55.
- [ABT19] Nathalie Aubrun, Sebastián Barbieri, and Stéphan Thomassé, *Realization of aperiodic subshifts and uniform densities in groups*, Groups Geom. Dyn. **13** (2019), no. 1, 107–129.
- [AS13] Nathalie Aubrun and Mathieu Sablik, *Simulation of effective subshifts by two-dimensional subshifts of finite type*, Acta Appl. Math. **126** (2013), 35–63.
- [Bar17] Sebastián Barbieri, *Shift spaces on groups : computability and dynamics*, Theses, Université de Lyon, June 2017.
- [Bar19] Sebastián Barbieri, *A geometric simulation theorem on direct products of finitely generated groups*, Discrete Analysis (2019), no. 9.
- [BBSS27] Sebastián Barbieri, Kanéda Blot, Mathieu Sablik, and Ville Salo, *A geometric obstruction to self-simulation for groups*, Discrete and Continuous Dynamical Systems **59** (2027), no. 0, 492–518.
- [BCVR24] Sebastián Barbieri, Nicanor Carrasco-Vargas, and Cristóbal Rojas, *Effective dynamical systems beyond dimension zero and factors of sfts*, Ergodic Theory and Dynamical Systems (2024), 1–41.
- [Ber66] Robert Berger, *The undecidability of the domino problem*, Mem. Amer. Math. Soc. **66** (1966).
- [BGv03] Laurent Bartholdi, Rostislav I. Grigorchuk, and Zoran Šuník, *Branch groups*, Handbook of Algebra, Vol. 3, Elsevier/North-Holland, Amsterdam, 2003, pp. 989–1112.
- [BM00] Marc Burger and Shahar Mozes, *Lattices in product of trees*, Publ. Math. Inst. Hautes Études Sci. **92** (2000), 151–194.
- [Boo59] William W. Boone, *The word problem*, Ann. of Math. (2) **70** (1959), 207–265.
- [Bri04] Matthew G. Brin, *Higher dimensional Thompson groups*, Geom. Dedicata **108** (2004), 163–192.
- [BS19] Sebastián Barbieri and Mathieu Sablik, *A generalization of the simulation theorem for semidirect products*, Ergodic Theory and Dynamical Systems **39** (2019), no. 12, 3185–3206.
- [BS26] Kanéda Blot and Ville Salo, *Self-simulability of graph products*, arXiv:2605.20945 (2026).
- [BSS25] Sebastián Barbieri, Mathieu Sablik, and Ville Salo, *Soficity of free extensions of effective subshifts*, Discrete Contin. Dyn. Syst. **45** (2025), no. 4, 1117–1149.
- [BSS26] ———, *Self-simulable groups*, Trans. Amer. Math. Soc. **379** (2026), no. 4, 2255–2306. MR 5039451

- [CFP96] James W. Cannon, William J. Floyd, and Walter R. Parry, *Introductory notes on Richard Thompson's groups*, Enseign. Math. (2) **42** (1996), no. 3-4, 215–256.
- [Cha07] Ruth Charney, *An introduction to right-angled Artin groups*, Geom. Dedicata **125** (2007), no. 1, 141–158.
- [Coh17] David Bruce Cohen, *The large scale geometry of strongly aperiodic subshifts of finite type*, Adv. Math. **308** (2017), 599–626.
- [CSC10] Tullio Ceccherini-Silberstein and Michel Coornaert, *Cellular automata and groups*, Springer Monographs in Mathematics, Springer, 2010.
- [Cul96] Karel Culik, II, *An aperiodic set of 13 Wang tiles*, Discrete Math. **160** (1996), no. 1-3, 245–251. MR 1417576
- [DRS12] Bruno Durand, Andrei Romashchenko, and Alexander Shen, *Fixed-point tile sets and their applications*, J. Comput. System Sci. **78** (2012), no. 3, 731–764.
- [Hed69] Gustav A. Hedlund, *Endomorphisms and automorphisms of the shift dynamical system*, Math. Systems Theory **3** (1969), 320–375.
- [Hig61] Graham Higman, *Subgroups of finitely presented groups*, Proc. Roy. Soc. London Ser. A **262** (1961), 455–475.
- [HM10] Michael Hochman and Tom Meyerovitch, *A characterization of the entropies of multidimensional shifts of finite type*, Ann. of Math. (2) **171** (2010), no. 3, 2011–2038.
- [Hoc09] Michael Hochman, *On the dynamics and recursive properties of multidimensional symbolic systems*, Invent. Math. **176** (2009), no. 1, 131–167.
- [Hoc16] ———, *Multidimensional shifts of finite type and sofic shifts*, Combinatorics, Words and Symbolic Dynamics, Encyclopedia of Mathematics and its Applications, vol. 159, Cambridge University Press, 2016, pp. 296–358.
- [JR21] Emmanuel Jeandel and Michaël Rao, *An aperiodic set of 11 Wang tiles*, Adv. Comb. (2021), Paper No. 1, 37. MR 4210631
- [Kar96] Jarkko Kari, *A small aperiodic set of Wang tiles*, Discrete Math. **160** (1996), 259–264.
- [LM21] Douglas Lind and Brian Marcus, *An introduction to symbolic dynamics and coding*, Cambridge Mathematical Library, Cambridge University Press, Cambridge, 2021, Second edition [of 1369092]. MR 4412543
- [LS77] Roger C. Lyndon and Paul E. Schupp, *Combinatorial group theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 89, Springer, 1977.
- [Nov55] Petr S. Novikov, *On the algorithmic unsolvability of the word problem in group theory*, Trudy Mat. Inst. Steklov. **44** (1955).
- [Pau91] Frédéric Paulin, *Outer automorphisms of hyperbolic groups and small actions on $\mathbb{R}$ -trees*, Arboreal Group Theory (Berkeley, CA, 1988), Math. Sci. Res. Inst. Publ., vol. 19, Springer, New York, 1991, pp. 331–343.
- [Rob71] Raphael M. Robinson, *Undecidability and nonperiodicity for tilings of the plane*, Invent. Math. **12** (1971), 177–209.

- [Rog67] Hartley Rogers, *Theory of recursive functions and effective computability*, McGraw-Hill, 1967.
- [Sew14] Brandon Seward, *Every action of a nonamenable group is the factor of a small action*, J. Mod. Dyn. **8** (2014), no. 2, 251–270.
- [Soa87] Robert I. Soare, *Recursively enumerable sets and degrees*, Perspectives in Mathematical Logic, Springer, 1987.
- [Tur36] Alan M. Turing, *On computable numbers, with an application to the Entscheidungsproblem*, Proc. London Math. Soc. (2) **42** (1936), 230–265.
- [Wan61] Hao Wang, *Proving theorems by pattern recognition II*, Bell System Tech. J. **40** (1961), 1–41.
- [Wei73] Benjamin Weiss, *Subshifts of finite type and sofic systems*, Monatsh. Math. **77** (1973), 462–474.
- [Why99] Kevin Whyte, *Amenability, bilipschitz equivalence, and the von Neumann conjecture*, Duke Math. J. **99** (1999), no. 1, 93–112.